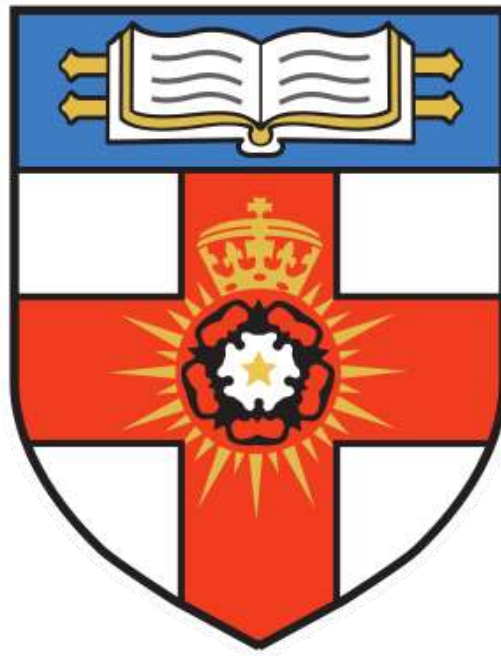


UNIVERSITY OF LONDON

INTERNATIONAL PROGRAMMES

BSc Computer Science and Related Subjects



CM3070 PROJECT

FINAL PROJECT REPORT

SenseFuzeAI: A Multimodal Behavioural Inference
Framework for Near-Real-Time Behavioural State
Classification

Author: Ng Jing Yi, Clara
Student Number : 240662088
Date of Submission: 28 September 2026
Supervisor : Mr. JL Chew
Word Count: 10413 / 10500

ABSTRACT

Behavioural-state inference from human–computer interaction is intrinsically multimodal: linguistic content, vocal characteristics, visual context, and interaction dynamics may provide complementary evidence, but they also introduce redundancy, dimensional imbalance, source-specific confounding, and uncertainty regarding behavioural construct validity. This project presents **SenseFuzeAI**, a modular multimodal behavioural-inference framework designed to investigate whether heterogeneous pretrained and engineered representations can be orchestrated for near-real-time classification of four operational behavioural states: **Focused, Distracted, Fatigued, and Overloaded**.

SenseFuzeAI integrates 22 engineered keystroke features with a 768-dimensional MPNet textual embedding, 809 audio features derived from WavLM and Librosa, and a 768-dimensional CLIP visual embedding. Together with six additional derived predictors, these form a 2,373-dimensional feature-level fusion representation. The implemented web-based architecture incorporates modality-specific preprocessing, deterministic record linkage, explicit feature-schema validation, behavioural abstraction of keystroke dynamics, supervised classification, confidence presentation, modality-status reporting, and prediction logging.

Evaluation was conducted on 309 near-balanced, class-aligned composite records using leakage-safe stratified five-fold cross-validation, supplemented by unimodal comparison, leave-one-modality-out retraining, inference-time zero-masking stress tests, computational-latency benchmarking, and automated software verification. Complete multimodal fusion achieved a mean macro-F1 of **1.0000**, compared with **0.9903** for the strongest unimodal keystroke model, representing an absolute improvement of **0.0097**. However, modality-removal experiments revealed that predictive contribution depended strongly on evaluation condition. Retrained ablation indicated substantial compensatory redundancy, whereas inference-time zero masking exposed greatest sensitivity to visual information, reducing macro-F1 to **0.9190**. Complete fusion further incurred substantially greater dimensionality and a mean warm processing latency of approximately **2.50 s**, demonstrating a measurable trade-off between predictive performance, computational cost, robustness, input availability, and interpretability.

These findings demonstrate the **engineering feasibility and within-dataset discriminative capability** of an inspectable multimodal orchestration framework, rather than participant-independent behavioural recognition. The experimental data comprise class-aligned rather than naturally co-recorded multimodal observations; keystroke evidence is participant-specific; audiovisual data have heterogeneous provenance; and the 2,373-predictor representation is evaluated on only 309 records. Consequently, the ceiling-level performance cannot establish external or psychological construct validity. SenseFuzeAI therefore contributes a reproducible architectural and methodological foundation for future research using naturally co-recorded multi-participant data, participant-aware validation, independent external evaluation, calibrated prediction, and more robust multimodal fusion.

Contents

ABSTRACT	 2-3
CHAPTER 1: INTRODUCTION.....	5-9
CHAPTER 2: LITERATURE REVIEW	10-21
CHAPTER 3: PROJECT DESIGN.....	22-39
CHAPTER 4: IMPLEMENTATION	40-52
CHAPTER 5: EVALUATION	53-83
CHAPTER 6: CONCLUSION	84-87
CHAPTER 7: APPENDICES	88-117
CHAPTER 8: REFERENCES.....	118-120

CHAPTER 1: INTRODUCTION (903/1000 words)

1.1 Project Concept and Problem Definition

Human behavioural state may be reflected through multiple, potentially complementary signals, including language, vocal characteristics, visual behaviour, and interaction patterns with digital systems. Although artificial intelligence has advanced substantially in natural-language processing, speech representation, computer vision, and behavioural biometrics, behavioural inference remains difficult when relevant evidence is distributed across heterogeneous modalities with differing dimensionalities, reliability, and semantic meaning.

This project presents SenseFuzeAI, a multimodal behavioural-inference framework that integrates keystroke, textual, audio, and visual information to classify a user's dominant behavioural state. Four operational classes are considered: **Focused**, **Distracted**, **Fatigued**, and **Overloaded**, representing sustained task engagement, attentional instability, reduced behavioural energy, and elevated task demand with interaction difficulty, respectively. These states are treated as mutually exclusive supervised-learning labels while recognising that the underlying conditions may overlap in real-world settings.

A distinguishing component is the use of keystroke dynamics as an interaction-level behavioural modality. Features such as dwell time, flight time, typing speed, timing variability, pausing, rhythm, and correction frequency provide observable evidence of interaction behaviour. Whereas keystroke dynamics have traditionally been used for authentication and user identification, SenseFuzeAI investigates whether such measurements provide complementary information for higher-level behavioural-state discrimination.

The project therefore addresses both classification and the systems-engineering problem of coordinating heterogeneous processors, maintaining record linkage, enforcing feature-schema consistency, integrating representations, and exposing interpretable outputs. The formal classification task and operational labels are defined in Section 3.4.

The evaluation is deliberately scoped as a within-dataset proof-of-concept. Because the experimental data comprise class-aligned composite records rather than naturally co-recorded behavioural episodes, and the keystroke component contains observations from one participant, the project does not claim population-level behavioural recognition, clinical diagnosis, or direct measurement of psychological state.

1.2 Motivation and Significance

Most digital systems respond primarily to explicit user input while possessing limited awareness of the behavioural conditions under which that input is produced. Behaviour-sensitive systems could support adaptive learning environments, human-computer interaction research, productivity applications, and other consent-based interfaces that respond to observable changes in engagement or cognitive demand.

Reliance on a single modality is potentially inadequate because behavioural signals are ambiguous and context-dependent. Semantically similar text may arise under different interaction conditions; irregular typing may reflect distraction, fatigue, planning, or task difficulty; acoustic features depend on recording conditions; and visual representations may capture environmental context as well as behaviourally relevant information. No individual signal should therefore be interpreted as a deterministic indicator of behavioural state.

Multimodal integration may provide complementary evidence, but additional modalities can also introduce redundancy, noise, high dimensionality, and spurious correlations. Robustness and predictive improvement must therefore be established empirically rather than assumed. SenseFuzeAI consequently investigates how heterogeneous evidence can be represented, record-aligned, fused, and evaluated within a modular and interpretable architecture.

1.3 Project Aim, Objectives, and Research Questions

The aim of this project is to design, implement, and evaluate SenseFuzeAI as a modular multimodal framework for near-real-time behavioural-state classification using heterogeneous AI representations and behavioural interaction features.

Near-real-time inference refers to producing a prediction immediately after processing a bounded multimodal interaction sample rather than performing continuous frame-level monitoring.

The project objectives are to:

1. investigate suitable feature-extraction approaches for keystroke, textual, audio, and visual data;
2. design a modular architecture for coordinating heterogeneous modality-specific processors;
3. develop a behavioural-abstraction layer that transforms low-level keystroke measurements into interpretable indicators;
4. implement a feature-level multimodal fusion pipeline for four-class behavioural-state classification; and
5. evaluate predictive performance, modality contribution, robustness, computational responsiveness, interpretability, and prototype limitations.

The project is guided by four research questions:

- **RQ1:** What software architecture can coordinate modality-specific models and record-aligned features to support near-real-time behavioural inference?
- **RQ2:** How does multimodal fusion compare with individual modalities in predictive performance and robustness?
- **RQ3:** How can low-level keystroke features be transformed into interpretable behavioural indicators, and what contribution do these indicators make to classification?
- **RQ4:** What trade-offs between predictive performance, computational responsiveness, robustness, and interpretability are observed in the implemented prototype?

1.4 Deliverables and Final Outcomes

The principal outcome is a functional web-based prototype that coordinates keystroke, textual, audio, and visual processing to produce a dominant behavioural-state prediction with supporting modality-level evidence.

The software deliverables comprise modality-specific processors, a keystroke behavioural-abstraction module, a multimodal fusion and classification pipeline, an integrated web application, persisted model artefacts, and an experimental evaluation framework. Supporting artefacts include source code, documentation, evaluation outputs, diagrams, and demonstration materials, with repository details provided in Appendix A.

Within its proof-of-concept scope, SenseFuzeAI makes three principal contributions:

1. a modular record-aligned orchestration architecture for heterogeneous behavioural modalities and pretrained representations;
2. an interpretable keystroke behavioural-abstraction layer that converts low-level interaction measurements into inspectable indicators without treating them as deterministic psychological measures; and
3. an empirical comparison of unimodal and multimodal behavioural-state classification, examining predictive performance, modality contribution, robustness, and computational responsiveness.

These contributions are interpreted as engineering and methodological contributions within the project dataset rather than evidence of a universally generalisable behavioural-state detector.

1.5 Project Template

This project is based on **CM3020 Artificial Intelligence – Project Idea 1: Orchestrating AI models to achieve a goal**. SenseFuzeAI addresses this project idea by integrating multiple AI models and modality-specific processing techniques within a unified behavioural-inference framework.

SenseFuzeAI satisfies this requirement by integrating textual-semantic, acoustic, visual, and keystroke interaction processing within a unified behavioural-inference architecture. The framework extends simple model aggregation through record-level alignment, explicit feature-schema validation, a custom keystroke abstraction layer, multimodal feature fusion, downstream classification, and presentation of modality-level evidence alongside the final prediction.

CHAPTER 2: LITERATURE REVIEW (2046/2500

words)

2.1 Domain Overview

Multimodal artificial intelligence concerns systems that learn from or reason across heterogeneous data sources. It is relevant to behavioural inference because engagement, attention, fatigue, and cognitive demand may be reflected through language, vocal characteristics, visual behaviour, and interaction patterns rather than a single observable signal.

Although advances in natural-language processing, speech representation, computer vision, and behavioural biometrics have made modality-specific feature extraction increasingly accessible, multimodal integration remains difficult because modalities differ in dimensionality, reliability, temporal structure, and semantic meaning. Effective behavioural inference therefore requires compatible representations, record alignment, an appropriate fusion strategy, and evaluation that distinguishes technical feasibility from behavioural validity.

This chapter reviews the multimodal, audio, visual, textual, keystroke, and fusion research informing SenseFuzeAI before synthesising the research gap addressed by the project.

2.2 Multimodal Artificial Intelligence

Baltrušaitis, Ahuja and Morency (2019) organise multimodal machine learning around five principal challenges: representation, translation, alignment, fusion, and co-learning. This taxonomy remains relevant because multimodality involves more than supplying several inputs to a classifier; heterogeneous observations must be represented consistently, aligned appropriately, and combined so that complementary, redundant, or conflicting evidence can be distinguished.

The principal motivation for multimodal learning is complementarity: different modalities may capture information that is unavailable or ambiguous in another. However, additional modalities do not necessarily provide additional unique information. Redundancy occurs when modalities encode overlapping discriminatory cues, meaning that a representation can perform strongly in isolation while contributing little incremental information after fusion. Conversely, a comparatively weak standalone modality may still provide complementary evidence for observations that another modality represents poorly. Multimodal benefit should therefore be established empirically rather than inferred from the number of available inputs.

Heterogeneous representations also introduce dimensional and reliability imbalance. High-dimensional pretrained embeddings may contribute substantially more predictors than compact engineered behavioural features without necessarily providing proportionally greater information. Under limited-data conditions, this imbalance can increase susceptibility to spurious correlations and overfitting, while differences in feature scale or reliability may cause particular modalities to dominate the learned decision boundary. Contemporary research consequently investigates uncertainty-aware and adaptive fusion rather than assuming equal contribution from every modality. For example, Gao et al. (2024) model unimodal aleatoric uncertainty, while He, Mi and Toda (2025) use gated interaction and modality-invariant learning.

Missing modalities constitute a related deployment challenge. Practical systems may encounter unavailable sensors, acquisition failures, privacy restrictions, or degraded observations (Wu et al., 2026). The existence of multiple modalities therefore does not itself establish robustness: performance must be examined when information is removed or becomes unreliable.

These considerations expose a complexity-data-interpretability trade-off. Highly parameterised fusion architectures may capture richer cross-modal interactions, but generally demand greater quantities of aligned data and can increase overfitting risk or reduce interpretability under constrained-data conditions. SenseFuzeAI therefore adopts a comparatively lightweight modular architecture combining fixed pretrained modality representations, engineered interaction features, and feature-level fusion. Multimodal superiority is not assumed; its incremental predictive value, redundancy, robustness, and computational consequences are treated as empirical questions developed further in Section 2.7 and evaluated in Chapter 5.

2.3 Audio-Based Behavioural Analysis

Speech contains multiple sources of potentially behaviour-relevant information, including prosody, speaking rate, pauses, pitch, energy, spectral characteristics, and temporal variation. Conventional acoustic analysis commonly represents these properties through engineered descriptors, while contemporary self-supervised speech models learn substantially higher-dimensional representations directly from large speech corpora.

WavLM is a self-supervised speech representation model designed to learn transferable information from raw audio and has demonstrated strong performance across diverse speech-processing tasks (Chen et al., 2022). SenseFuzeAI combines a 768-dimensional WavLM representation with 41 Librosa-derived acoustic descriptors. This hybrid approach provides both a broad pretrained representation and explicit signal-level measurements rather than relying exclusively on either representation type.

However, acoustic representation capability must be distinguished from behavioural construct validity. WavLM embeddings are not dedicated measurements of fatigue, distraction, overload, or attention; they may simultaneously encode linguistic content, speaker characteristics, vocal delivery, recording environment, and other source-specific information. Likewise, engineered descriptors such as pitch or energy may vary with behavioural condition but are also affected by speaker physiology, microphone characteristics, speaking style, and recording context.

Strong acoustic classification can demonstrate separability without establishing which acoustic factors produced it or whether the relationship generalises across speakers and recording environments. SenseFuzeAI therefore treats WavLM and Librosa features as complementary evidence rather than direct behavioural-state measurements.

2.4 Vision-Based Behavioural Analysis

Visual behavioural analysis commonly uses facial expression, head pose, gaze direction, and eye-related cues. OpenFace 2.0 provides explicit facial landmarks, head pose, gaze estimates, and facial action units, making it suitable when behaviourally interpretable facial measurements are required (Baltrušaitis et al., 2018).

CLIP follows a different approach. Rather than extracting predefined behavioural measurements, it learns transferable visual-semantic representations from large-scale image-text supervision (Radford et al., 2021). Such representations can encode broad information about appearance, objects, pose, scene, and context, providing flexibility when visual evidence extends beyond facial geometry.

The trade-off is therefore representational breadth versus construct-specific interpretability. A CLIP representation may discriminate experimental classes effectively while relying on contextual or source-specific visual regularities rather than behaviourally meaningful facial cues. Strong classification performance consequently demonstrates visual-semantic separability, not that CLIP directly measures fatigue, distraction, overload, or attention.

SenseFuzeAI uses CLIP because its visual inputs are heterogeneous rather than restricted to controlled facial imagery, favouring broad transferable representation over the explicit behavioural measurements available from systems such as OpenFace.

2.5 Text-Based Semantic Analysis

Transformer-based language models provide contextual representations that capture semantic relationships beyond surface-level lexical features. MPNet combines masked and permuted language-modelling objectives to produce contextual language representations (Song et al., 2020), while Sentence-BERT-style architectures enable efficient fixed-dimensional sentence embeddings suitable for downstream similarity and classification tasks (Reimers and Gurevych, 2019).

SenseFuzeAI uses the 768-dimensional all-mpnet-base-v2 sentence embedding as a fixed semantic representation. This avoids training a large language model on the comparatively small project dataset and provides a consistent representation for integration with other modalities.

However, semantic separability must be distinguished from construct validity. Experimental classes may be separable through vocabulary, topic, phrasing, or class-guided construction without showing that the embedding measures the underlying behavioural state. High text-classification performance therefore demonstrates discriminable semantic representations, not independent validation of Focused, Distracted, Fatigued, or Overloaded as psychological constructs.

Text is consequently treated as complementary evidence whose incremental contribution is evaluated against the other modalities.

2.6 Keystroke Dynamics and Behavioural Biometrics

Keystroke dynamics characterise keyboard interaction through features such as dwell time, flight time, typing speed, pauses, corrections, and temporal variability, as illustrated in Figure 2.1. Unlike camera- or microphone-based sensing, these measurements can be captured through ordinary computer interaction without additional physical sensors.

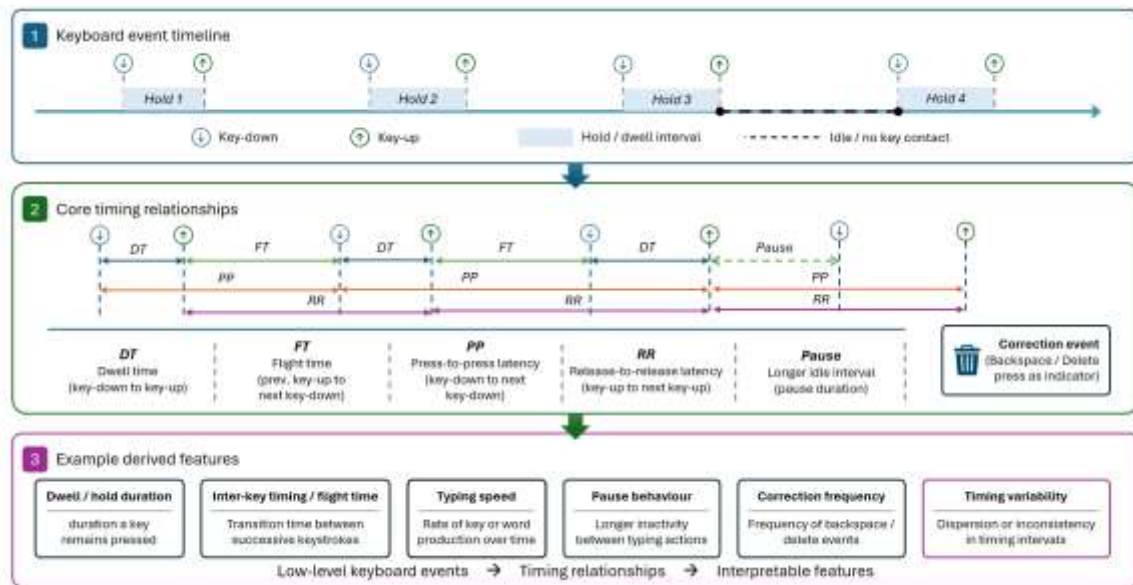


Figure 2.1. Keystroke timing relationships used in keystroke-dynamics feature extraction.

Historically, keystroke dynamics have been studied primarily as a behavioural biometric for authentication and user identification. Contemporary approaches range from engineered timing features to deep sequential models, reflecting the extent to which typing behaviour contains stable individual characteristics (Shadman et al., 2025). This stability is advantageous for identity recognition but creates an important methodological problem for behavioural-state inference: features intended to capture temporary state variation may simultaneously encode persistent participant-specific motor patterns.

This produces a fundamental state-versus-identity confound: high within-participant classification may reflect temporary condition-associated changes, stable motor characteristics, or interactions between both. Participant-independent claims therefore require training and testing across separated participant identities. Because SenseFuzeAI's principal keystroke evaluation involves one participant, its evidence supports within-participant discrimination rather than generalisation across users.

Research nevertheless indicates that keystroke behaviour can vary with transient cognitive and affective conditions. Epp, Lippold and Mandryk (2011) reported relationships between keystroke features and self-reported emotional states, while Acien et al. (2022) investigated keystroke dynamics as a potential real-world marker of mental fatigue. Bennett, Roudaut and Metatla (2025) further associated fatigue with changes in the temporal organisation of typing. More recently, Schütz et al. (2026) found that increased task difficulty during human-LLM interaction was associated with slower typing and greater pausing, although keystroke measures did not predict every construct examined.

Taken together, these studies support keystrokes as potentially informative behavioural evidence while demonstrating that their interpretation is construct- and context-dependent. Pausing may reflect distraction, fatigue, planning, or task difficulty; correction frequency may indicate overload but can also arise from unfamiliar vocabulary or typing errors. Device type, language, typing proficiency, task structure, and individual motor habits introduce further variation. Consequently, no single keystroke feature should be treated as a deterministic indicator of behavioural state.

SenseFuzeAI groups low-level measurements into inspectable indicators of rhythm, pausing, correction activity, speed, and variability. These are operational rather than causal psychological descriptors and are evaluated both standalone and within multimodal ablation.

2.7 Multimodal Fusion and AI Orchestration

Fusion strategies are commonly distinguished by the stage at which modality information is combined. Feature-level fusion concatenates or integrates representations before classification, intermediate fusion learns shared cross-modal representations within the model, and decision-level fusion combines predictions from separately trained modality-specific classifiers (Baltrušaitis, Ahuja and Morency, 2019), as summarised in Figure 2.2.

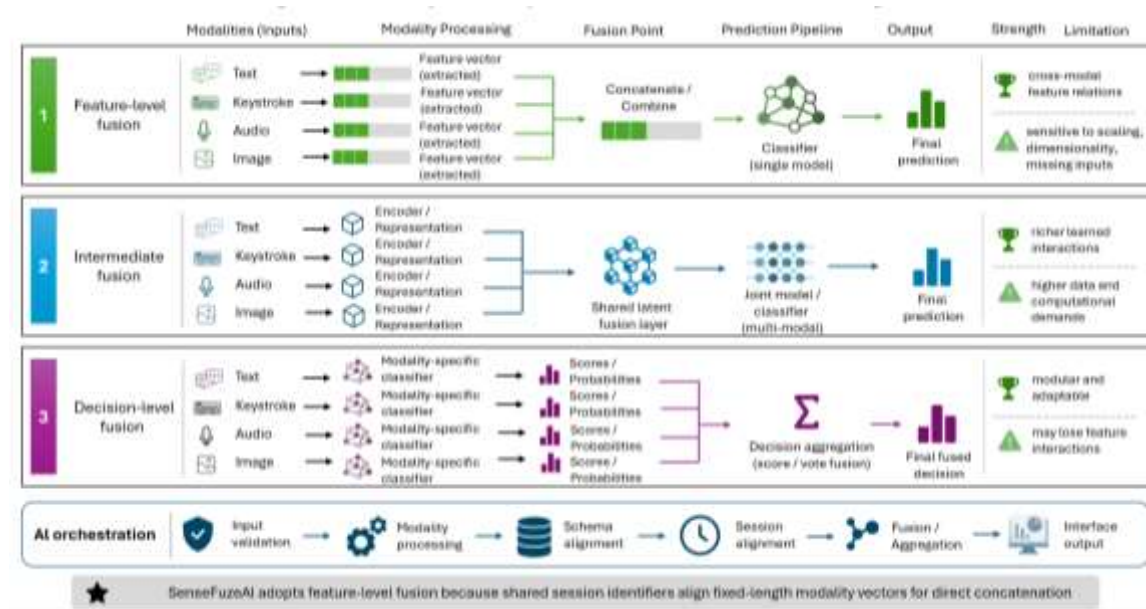


Figure 2.2. Conceptual comparison of multimodal fusion strategies. Author synthesis based on Baltrušaitis et al. (2019).

Feature-level fusion provides a direct mechanism for modelling relationships among heterogeneous representations and enables consistent comparison between unimodal and multimodal configurations. Its principal limitations include dimensional imbalance, incompatible feature scales, redundancy, and dependence on reliable record alignment. Decision-level fusion offers stronger modularity and can accommodate unavailable modalities more naturally, but may discard low-level cross-modal interactions and depends on meaningful modality-specific classifier outputs. Intermediate approaches can capture richer interactions, although this generally increases model complexity, computational demand, and data requirements.

Fusion strategy should be selected not only for predictive capacity but also for sample size, modality availability, interpretability, and deployment constraints. Under constrained-data conditions, greater architectural sophistication may increase representational capacity without improving reliable generalisation.

Contemporary work increasingly models unequal modality reliability through uncertainty-aware or adaptive mechanisms. Gao et al. (2024), Huang et al. (2025), He, Mi and Toda (2025), and Dutta, Balaji and Ganapathy (2026) illustrate approaches in which modality influence is adjusted rather than assumed equal.

A further distinction concerns redundancy versus robustness. A modality may appear redundant after retraining because the remaining representations compensate, yet a classifier trained on the complete feature distribution may still depend on it. Missing-modality evaluation should therefore distinguish retrained ablation from inference-time modality loss.

SenseFuzeAI deliberately adopts feature-level fusion to provide an inspectable basis for unimodal-multimodal comparison while limiting additional trainable complexity. Orchestration additionally requires record linkage, schema consistency, failure handling, classifier execution, and presentation; evaluation therefore separates predictive contribution, modality-loss robustness, and operational responsiveness.

Area or approach	Principal strength	Principal limitation	Design Implication
Multimodal-learning taxonomy	Identifies representation, alignment, and fusion as distinct problems	Does not prescribe one architecture for all applications	Separate modality processing, alignment, and fusion responsibilities
MPNet-based sentence embeddings	Compact contextual semantic representation	May encode task wording rather than behavioural condition	Compare text-only and fused performance
WavLM audio representations	Captures broad speech and acoustic information	Features are not intrinsically behaviourally interpretable	Treat audio as probabilistic complementary evidence

CLIP visual embeddings	Provides transferable visual-semantic representations	Not specialised for gaze, fatigue, or workload measurement	Use as contextual visual evidence
OpenFace-style visual analysis	Produces interpretable facial and gaze measurements	Sensitive to visibility, pose, and capture conditions	Relevant alternative for future specialised visual analysis
Engineered keystroke features	Continuous, low-cost interaction evidence	Strongly participant-, task-, and device-dependent	Use behavioural abstraction, evaluate both standalone and incremental contribution through ablation, and restrict generalisation claims
Feature-level fusion	Supports cross-modal feature relationships	Vulnerable to dimensionality, redundancy, scaling, and missing inputs	Apply schema alignment and normalisation, compare unimodal and fused performance, and evaluate modality contribution and missing-modality robustness empirically
Decision-level fusion	Modular and adaptable to unavailable modalities	May lose feature interactions and requires calibrated outputs	Appropriate comparative or future extension

Table 2.1. Comparative synthesis of literature informing SenseFuzeAI

2.8 Critical Synthesis and Research Gap

The reviewed literature reveals three central conclusions. First, each modality can provide potentially useful behavioural evidence, but none directly measures the operational states used in this project. Text may encode task wording; audio and vision may capture speaker, source, or environmental characteristics; and keystrokes are strongly influenced by participant, task, device, and context. Behavioural interpretation must therefore remain probabilistic and dataset-specific rather than being treated as direct measurement of latent psychological state.

Second, pretrained representations such as MPNet, WavLM, and CLIP provide powerful generic feature spaces, but they were not developed as validated detectors of Focused, Distracted, Fatigued, or Overloaded. Representation capability must therefore be distinguished from construct validity. High classification performance may demonstrate separability of experimentally labelled observations without establishing that the representation measures the intended behavioural construct.

Third, contemporary multimodal research already demonstrates sophisticated uncertainty-aware, attention-based, and expert-gated fusion. The research gap addressed by SenseFuzeAI is therefore not the absence of multimodal fusion methods. Rather, it concerns whether a comparatively lightweight and inspectable architecture can coordinate heterogeneous pretrained representations with interpretable keystroke-derived evidence under constrained-data conditions, while exposing the trade-offs between complementary information, redundancy, missing-modality sensitivity, computational cost, and interpretability.

Within this scope, the contribution lies in combining record-aligned multimodal orchestration, interpretable keystroke abstraction, and controlled unimodal-multimodal evaluation within one deployment-oriented framework, rather than proposing a new foundation model or claiming universal behavioural-state recognition.

These gaps motivate the project's four research questions. RQ1 addresses how heterogeneous pretrained and engineered representations can be coordinated within a record-aligned architecture. RQ2 tests whether fusion provides incremental predictive value and robustness rather than assuming multimodal superiority. RQ3 examines whether keystroke measurements can provide interpretable behavioural evidence while recognising participant dependence. RQ4 evaluates whether any predictive advantage justifies the associated computational, robustness, and interpretability costs. Chapter 3 translates these literature-derived requirements into the system design.

CHAPTER 3: PROJECT DESIGN (1937/2000 words)

3.1 Design Overview

SenseFuzeAI was designed to address the integration challenges identified in Chapter 2: heterogeneous feature representations, modality alignment, interpretability, processing latency, and the risk of dependence on a single behavioural signal. Rather than employing a monolithic end-to-end neural architecture, the system adopts a modular pipeline in which keystroke, textual, audio, and visual inputs are processed independently before their feature representations are aligned and combined.

The architecture follows a systems-engineering approach. Each modality-specific component has a defined input, preprocessing procedure, feature schema, and output interface. Shared identifiers provide deterministic record-level linkage between modality-specific observations, enabling schema-consistent composite records to be constructed for training and evaluation. This linkage does not imply that the modalities were acquired simultaneously or originated from the same participant or behavioural episode.

Early development explored heuristic mappings and manually calibrated behavioural scores. These prototypes supported feasibility analysis but became difficult to justify and maintain as additional modalities were introduced. The final design therefore replaced rule-dominated classification with a data-driven supervised-learning pipeline based on record-linked, class-aligned observations. The heuristic work was retained only where it informed interpretable keystroke indicators and interface explanations.

3.2 Domain, Users, and Design Principles

SenseFuzeAI is positioned within multimodal artificial intelligence, behavioural computing, and human-computer interaction. Its immediate purpose is experimental: it provides a platform for investigating whether heterogeneous interaction signals can support classification of four operational behavioural states. It is not designed as a clinical diagnostic system or as a covert monitoring application.

The principal users are researchers, developers, and evaluators examining multimodal model orchestration and behavioural interaction data. Their requirements include access to modality-specific outputs, traceability of predictions, repeatable processing, and the ability to replace individual components without redesigning the entire system.

Five principles guided the design:

1. **Modularity:** modality processors and classification components should remain independently replaceable.
2. **Separation of concerns:** acquisition, preprocessing, feature extraction, fusion, classification, and presentation should have distinct responsibilities.
3. **Interpretability:** predictions should be accompanied by inspectable modality-level evidence and keystroke indicators.
4. **Reproducibility:** training and inference should use explicit feature schemas, stored model artefacts, and consistent preprocessing.
5. **Practical responsiveness:** bounded multimodal samples should be processed with sufficiently low latency for interactive use.

These principles prioritise experimental transparency and maintainability over architectural complexity.

3.3 Requirements

The system requirements are divided into Functional Requirements (FRs) and Non-Functional Requirements (NFRs). Functional requirements define the behaviours and capabilities that the prototype must provide, while non-functional requirements specify the quality attributes and operational characteristics expected of the system. The identifiers FR1-FR8 and NFR1-NFR8 are used throughout this section for traceability.

The functional requirements are summarised in Table 3.1.

Requirement ID	Functional Requirement	Rationale
FR1	Acquire multimodal behavioural data from text, audio, image, and keystroke inputs.	Supports heterogeneous behavioural evidence collection.
FR2	Perform modality-specific preprocessing and feature extraction.	Ensures each modality is represented in a suitable format for downstream analysis.
FR3	Execute specialised AI models independently for each behavioural modality.	Enables modular processing and

		independent model optimisation.
FR4	Transform low-level behavioural observations into interpretable behavioural indicators through the behavioural abstraction layer.	Improves behavioural interpretability prior to multimodal fusion.
FR5	Integrate modality-specific outputs using a multimodal fusion mechanism.	Produces a unified behavioural representation from heterogeneous evidence.
FR6	Classify behavioural state as Focused, Distracted, Fatigued, or Overloaded.	Supports the behavioural inference objective of the project.
FR7	Present behavioural predictions and confidence information through the web interface.	Enables users to interpret behavioural outcomes.
FR8	Maintain modular separation between modality-specific components.	Supports maintainability and future extensibility.

Table 3.1. Functional Requirements

The non-functional requirements are summarised in Table 3.2 and define the principal quality attributes and operational constraints of the prototype.

Requirement ID	Non-Functional Requirement	Quality Criterion / Design Consideration
NFR1	Near-real-time performance	Behavioural predictions should be generated with sufficiently low latency to support responsive interaction.
NFR2	Modularity	Individual AI models should be independently

		replaceable without affecting the overall architecture.
NFR3	Scalability	The framework should support additional behavioural modalities and future AI models.
NFR4	Extensibility	New behavioural features and fusion strategies should be incorporated with minimal architectural modification.
NFR5	Robustness	The system should continue operating when one modality is unavailable or degraded.
NFR6	Interpretability	Behavioural predictions should be traceable to modality-level behavioural evidence.
NFR7	Maintainability	Software components should remain logically organised and straightforward to update.
NFR8	Usability	The web interface should provide clear behavioural outputs and intuitive interaction for users.

Table 3.2. Non-functional requirements

3.4 Classification Problem and Operational Labels

SenseFuzeAI formulates behavioural inference as a supervised four-class classification problem:

$$\hat{b}^{(i)} = f_{\theta}(\mathbf{x}_K^{(i)}, \mathbf{x}_T^{(i)}, \mathbf{x}_A^{(i)}, \mathbf{x}_V^{(i)}, \mathbf{x}_D^{(i)})$$

where $\mathbf{x}_K^{(i)}$, $\mathbf{x}_T^{(i)}$, $\mathbf{x}_A^{(i)}$, and $\mathbf{x}_V^{(i)}$ denote the keystroke, textual, audio, and visual feature vectors for record i , respectively; $\mathbf{x}_D^{(i)} \in R^6$ denotes the six additional derived predictors retained in the final feature schema; f_{θ} is the learned classifier; and $\hat{b}^{(i)}$ is the predicted behavioural-state label:

$$\mathcal{B} = \{\text{focused, distracted, fatigued, overloaded}\}$$

These labels represent operational dominant states defined for controlled experimental classification rather than discrete or clinically validated psychological conditions.

State	Operational definition	Indicative behavioural evidence
Focused	Sustained task engagement with comparatively stable interaction behaviour	Consistent typing rhythm, limited pausing, task-relevant language, and sustained visual attention
Distracted	Repeated diversion of attention away from the primary task	Irregular interaction timing, interruption-related pauses, environmental audio activity, or visual disengagement
Fatigued	Reduced behavioural energy or responsiveness during task performance	Slower interaction, prolonged pauses, reduced vocal energy, and less consistent visual engagement

Overloaded	Elevated task demand accompanied by increased interaction difficulty or instability	Increased correction behaviour, variable timing, hesitation, strained vocal characteristics, or disrupted task progression
------------	---	--

Table 3.3. Operational definitions and illustrative indicators for the four behavioural-state classes

Table 3.3 presents design hypotheses rather than deterministic rules. Each modality contributes complementary evidence, while the keystroke-abstraction layer groups low-level interaction measures into inspectable indicators. These indicators are not direct or causal measurements of psychological state.

3.5 Overall Architecture

Figure 3.1 presents the SenseFuzeAI architecture as six connected processing layers:

1. multimodal data acquisition;
2. modality-specific preprocessing;
3. feature extraction;
4. record linkage and feature-schema validation;
5. feature-level fusion and classification;
6. presentation and explanation.

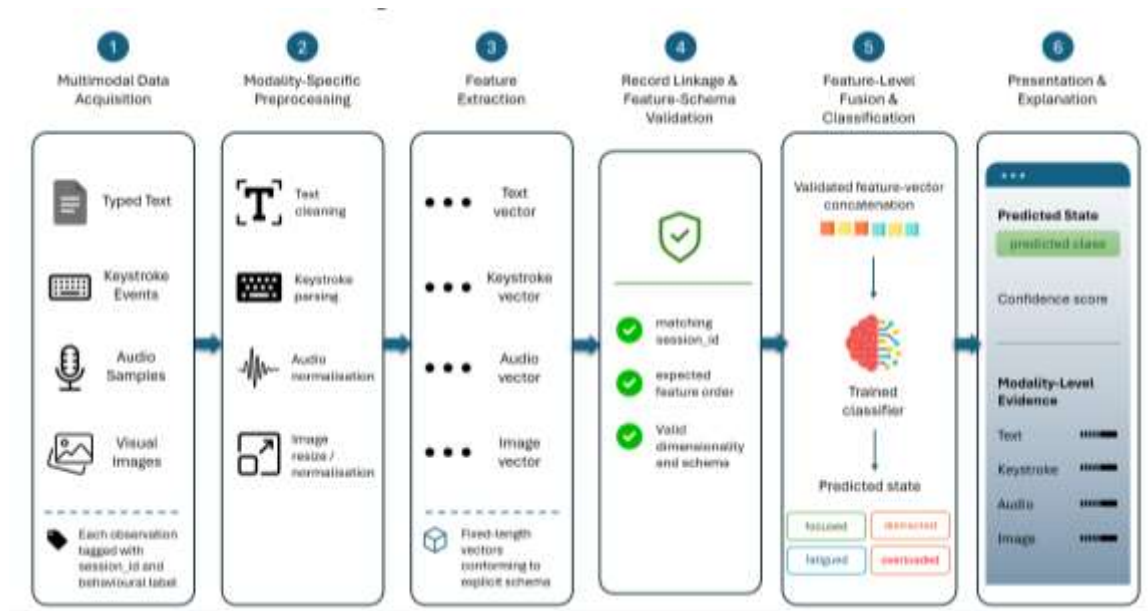


Figure 3.1. Overall SenseFuzeAI architecture.

The acquisition layer supports typed text, keystroke events, audio samples, and visual images. Each stored observation carries a record identifier and operational behavioural label, allowing modality-specific feature tables to be linked consistently during dataset construction. For the final evaluation dataset, these identifiers support deterministic record linkage rather than temporal synchronisation.

The preprocessing and feature-extraction layers transform raw inputs into numerical representations. Because the modalities differ substantially, they are not forced through a common preprocessing procedure. Each processor instead produces a fixed-length vector conforming to an explicit schema.

The record-linkage layer verifies that modality-specific observations carry the expected shared identifier and class association, and that their feature vectors match the required ordering and dimensionality. For the constructed evaluation dataset, this establishes deterministic schema-consistent linkage between class-aligned records; it does not establish that the underlying signals were captured simultaneously or during the same natural behavioural episode.

At the fusion layer, modality-specific representations are combined with the additional derived predictors using an explicit persisted feature schema. The final classifier input therefore contains 2,373 predictive dimensions: 22 keystroke features, 768 textual features, 809 audio features, 768 visual features, and six additional derived predictors.

The fusion stage validates both column ordering and expected dimensionality before classification. This is important because the classifier was trained against a fixed 2,373-dimensional schema; inconsistent ordering, omitted predictors, or unexpected dimensions could otherwise produce invalid or irreproducible predictions.

The presentation layer exposes the prediction through the web interface together with confidence information and selected modality-level evidence. This layered architecture allows failures to be localised and individual processors to be replaced without modifying the entire pipeline.

3.6 Modality-Specific Design

3.6.1 Text Analysis Module

The text module transforms typed content into an MPNet-based semantic embedding. The design uses a pretrained representation rather than training a language model from the project dataset, which would be unsuitable given its scale.

The embedding provides a fixed-length representation for classification. It does not directly encode focus, fatigue, distraction, or overload; these associations are learned by the downstream classifier. Text preprocessing and embedding generation are therefore applied identically during dataset construction and application inference.

3.6.2 Keystroke Analysis Module

The keystroke module captures key-down and key-up events during text entry. From these events, it derives interaction features including dwell time, flight time, typing speed, pause behaviour, correction frequency, and timing variability.

Keystroke analysis is separated from textual analysis because it represents how the user interacts rather than what the user writes. This distinction enables semantically similar text samples to exhibit different interaction-level patterns.

The module also generates higher-level indicators relating to rhythm stability, pausing, correction behaviour, and interaction speed. These indicators support interpretation but are not treated as deterministic rules for assigning behavioural labels.

3.6.3 Audio Analysis Module

The audio module converts a bounded recording into a WavLM-based feature representation. WavLM was selected to provide a general representation of speech and acoustic characteristics without requiring a task-specific audio network to be trained from first principles.

The module validates the audio format, applies the required sampling and preprocessing operations, and generates a fixed-length vector. Because vocal characteristics may reflect speaker identity or recording conditions, the audio representation is treated as complementary evidence rather than as an independently conclusive measure.

3.6.4 Vision Analysis Module

The visual module uses CLIP to obtain a semantic representation of an uploaded or captured image. This provides broader contextual information than manually selected image statistics and produces a fixed-length vector suitable for fusion.

A webcam-domain calibration stage was subsequently incorporated to reduce representation mismatch between the original image sources and images encountered during live webcam operation.

CLIP is not treated as a specialised detector of gaze, fatigue, or cognitive workload. Its output is instead used as contextual visual evidence whose value is established empirically through unimodal and multimodal comparisons.

3.7 Behavioural Abstraction and Fusion

The behavioural-abstraction component is applied principally to keystroke dynamics. Low-level timing measurements are difficult to interpret individually and may have several plausible explanations. The abstraction layer therefore groups selected measurements into inspectable indicators such as rhythm stability, pausing intensity, correction activity, and variation in interaction speed.

These indicators serve two purposes. First, they expose evidence that can be presented alongside the final prediction. Second, they provide engineered behavioural representations that can be evaluated together with the underlying statistical keystroke features. They do not replace supervised learning and are not assumed to represent universal psychological relationships.

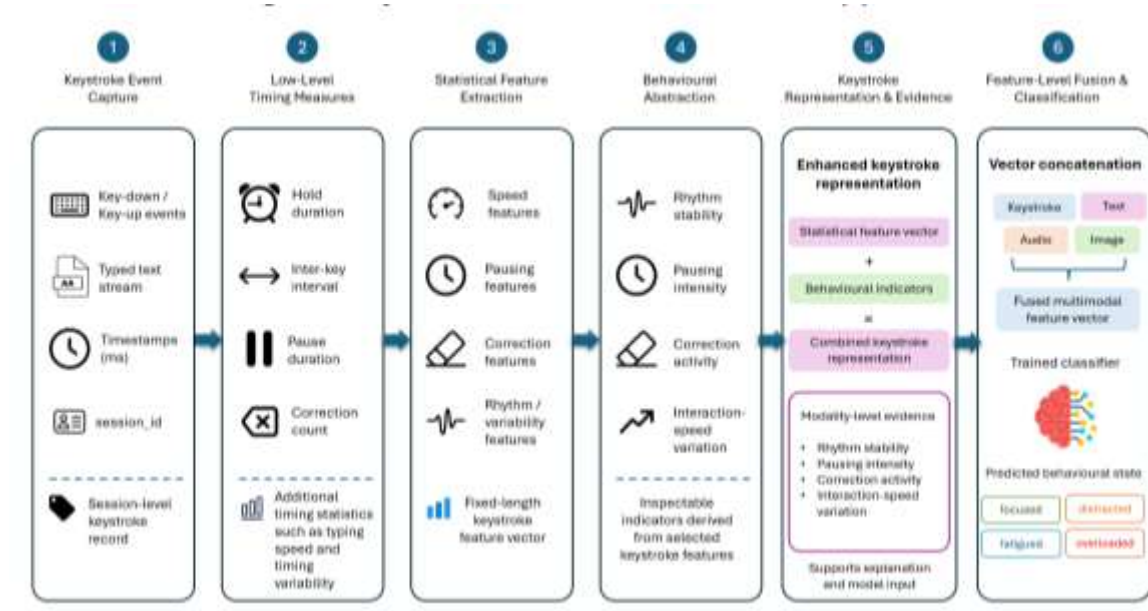


Figure 3.2. Keystroke feature and behavioural-abstraction pipeline.

Multimodal integration is performed using feature-level fusion. After each modality-specific vector has been generated, validated, and arranged according to its defined feature schema, the representations are concatenated in a predetermined order:

$$x_F = x_K \parallel x_T \parallel x_A \parallel x_V$$

where x_K , x_T , x_A , and x_V denote the keystroke, textual, audio, and visual feature vectors, respectively; $\parallel$ denotes vector concatenation; and x_F denotes the resulting fused feature vector.

The fused representation is subsequently supplied to a trained classifier:

$$\hat{b} = g_{\theta}(x_F)$$

where g_{θ} denotes the classifier with learned parameters θ , and $\hat{b}$ is the predicted dominant behavioural state.

Feature-level fusion was selected because the modality observations are associated at record level through shared identifiers and represented as fixed-length vectors. This enables construction of a consistent joint feature space for experimental comparison, while recognising that record-level association does not constitute natural temporal synchronisation. However, the approach also introduces risks relating to dimensional imbalance, feature scaling, overfitting, and incomplete inputs. These risks are addressed through explicit feature-schema validation, consistent preprocessing, comparative unimodal baselines, and evaluation of input-availability constraints.

3.8 User Interface Design

The interface was designed as a web-based dashboard through which users can provide multimodal inputs and inspect the resulting prediction. It includes a text-entry area with keystroke capture, audio and image input controls, processing feedback, and a results panel.

The results panel presents the predicted dominant state, confidence information, and selected supporting indicators. The interface deliberately avoids presenting the output as a clinical or definitive judgement. Instead, it communicates the result as a model prediction produced from the available interaction sample.

Usability priorities include clear input states, validation messages, predictable navigation, and separation between data acquisition and result presentation.

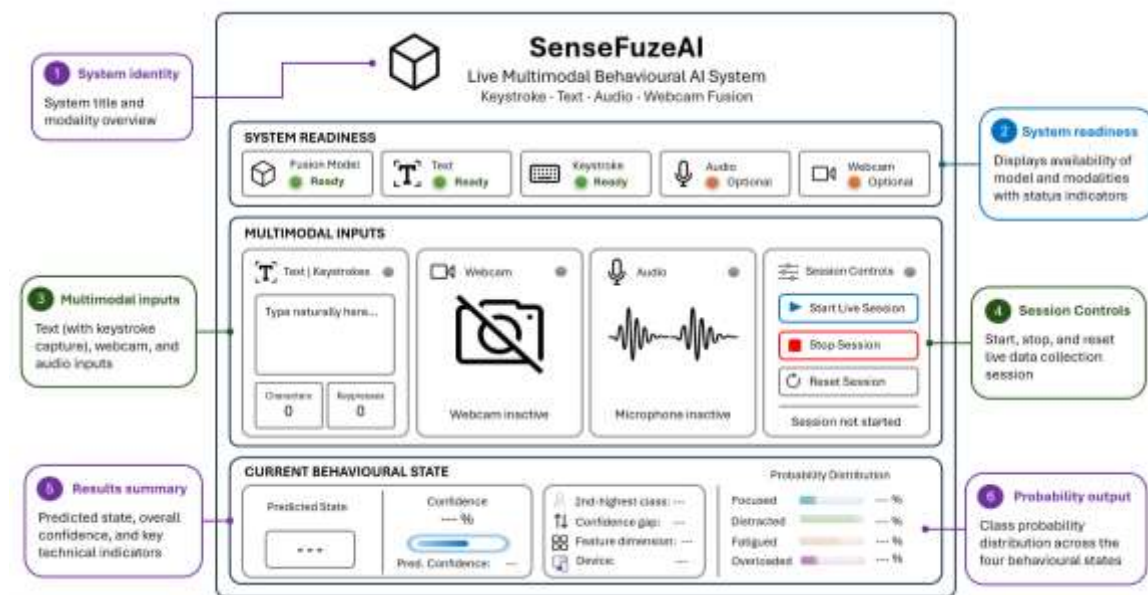


Figure 3.3. SenseFuzeAI interface design.

3.9 Technology Selection

Technology selection was based on compatibility with Python machine-learning libraries, modular integration, reproducibility, and browser accessibility.

Technology	Role	Selection rationale
Python	Core processing and experimentation	Broad machine-learning and scientific-computing support
FastAPI	Back-end application and inference endpoints	Lightweight Python integration and structured request handling
HTML, CSS, JavaScript and Bootstrap	Browser interface	Accessible multimodal input and responsive presentation
MPNet-based sentence transformer	Text representation	Fixed-length contextual semantic embeddings
WavLM	Audio representation	Pretrained speech and acoustic feature extraction
CLIP	Visual representation	Transferable image-semantic embeddings
NumPy and Pandas	Feature processing	Structured numerical and tabular operations
Scikit-learn	Classification and evaluation	Reproducible conventional classifiers and metrics
Matplotlib and Seaborn	Evaluation visualisation	Confusion matrices and comparative result figures

Table 3.4. Principal technology choices

Scikit-learn, PyTorch, and FastAPI were selected for reproducibility, compatibility with the pretrained feature extractors, and straightforward integration into the prototype architecture rather than claimed superiority over alternatives.

3.10 Testing and Evaluation Strategy

The design distinguishes software verification from experimental model evaluation.

Software verification comprises:

- unit testing of isolated utilities, prediction logic, and temporal aggregation;
- integration testing of model artefacts, feature schemas, calibration components, and inference modules;
- system testing of FastAPI endpoints, browser-facing controls, request validation, and temporal-state behaviour;
- acceptance and smoke/regression testing against functional requirements and implementation changes; and
- dataset-comparison verification of keystroke harmonisation, provenance, fixed split contracts, leakage prevention, and comparison artefacts.

Level	Purpose
Unit	Verify isolated utility, input-handling, probability, and temporal-aggregation logic
Integration	Verify model artefacts, calibration components, feature-schema compatibility, and inference integration
System	Validate FastAPI endpoints, request handling, browser-facing controls, and temporal-state behaviour
Acceptance	Assess compliance with final functional requirements and required project artefacts
Smoke / Regression	Detect cross-component failures following implementation changes
Dataset-comparison verification	Verify keystroke feature harmonisation, provenance, participant/session

	separation, frozen splits, leakage prevention, and comparison-result contracts
--	--

Table 3.5. Software-testing strategy

Experimental evaluation uses stratified five-fold cross-validation, accuracy, macro-precision, macro-recall, macro-F1, and confusion matrices. Additional evaluation includes leave-one-modality-out ablation, simulated missing-modality stress testing, repeated CPU-based latency benchmarking, webcam-domain adaptation, and interpretability analysis; the separate keystroke dataset-comparison pipeline is software-verified in Section 5.8.

Because the principal keystroke dataset contains observations from one participant, the primary cross-validation results measure within-dataset discrimination rather than generalisation to unseen users. Chapter 5 reports the resulting evidence and limitations.

3.11 Development Plan

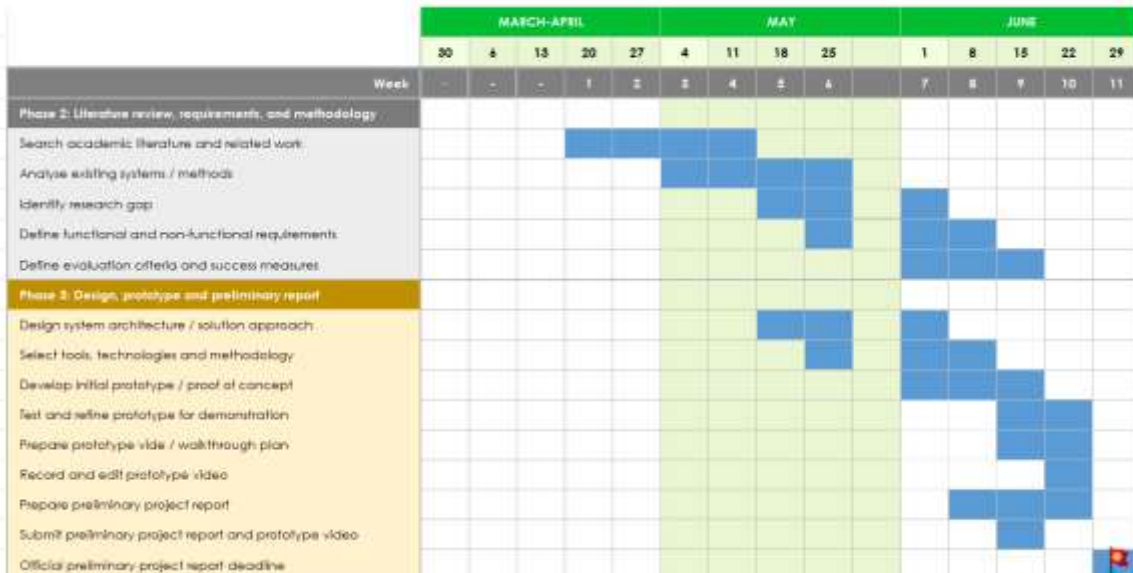
Development proceeded iteratively from literature review and prototyping through architecture refinement, record-linked data construction, implementation, integration, testing, evaluation, and reporting. Heuristic scoring was replaced by supervised classification once the rule-based approach proved difficult to justify and scale.

Figure 3.4 summarises the project schedule and shows the dependencies between dataset construction, model development, application integration, evaluation, and report completion.

CM3070 Final Project
Project Workplan and Gantt Chart Timeline (April–September 2024)
Student UOL ID: 240662088



CM3070 Final Project
Project Workplan and Gantt Chart Timeline (April–September 2024)
Student UOL ID: 240662088



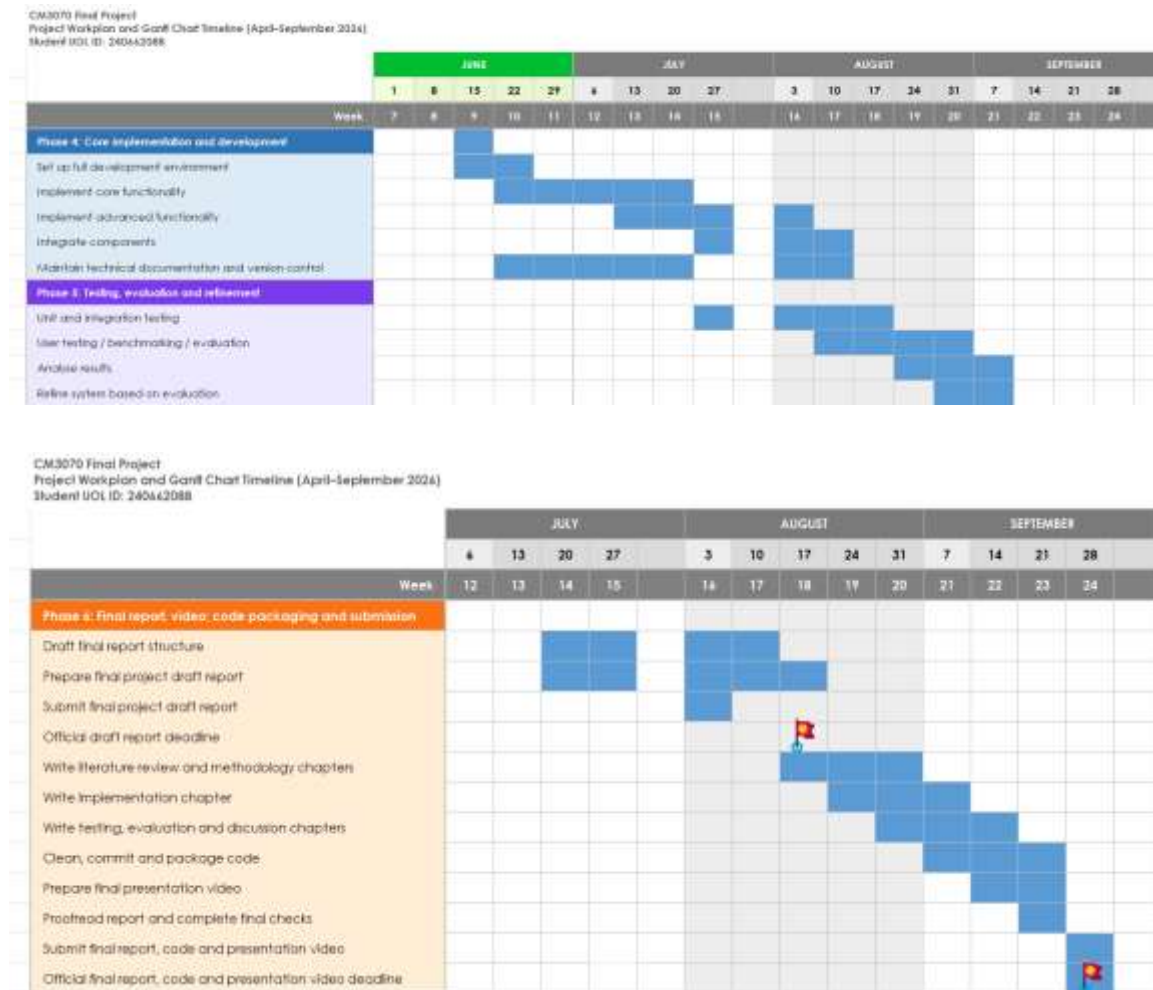


Figure 3.4. Project development schedule.

This iterative process allowed design decisions to be revised in response to implementation evidence while preserving traceability between the research objectives, system requirements, and final architecture.

CHAPTER 4: IMPLEMENTATION (2196/2500 words)

4.1 Implementation Overview

SenseFuzeAI was implemented as a modular client-server application coordinating keystroke, text, audio, and visual processing across acquisition, feature extraction, record-linked integration, and supervised classification. The system uses a Python/FastAPI backend and browser interface, with modality processors separated from the web layer so extraction, training, and deployment can be tested independently.

The final pipeline combines engineered keystroke features, MPNet text embeddings, WavLM plus acoustic descriptors, and CLIP visual embeddings under a persisted feature schema. Early heuristic scoring prototypes were replaced by supervised classifiers because the rule-based approach became difficult to justify and scale.

4.2 Development Environment and Reproducibility

4.2.1 Development Environment

The backend was developed in Python 3.11.9 with FastAPI for request handling and inference endpoints, Jinja2 and JavaScript for the browser interface, Scikit-learn for conventional machine learning, PyTorch for pretrained feature extractors, Librosa for engineered acoustic descriptors, and NumPy/Pandas for numerical and tabular processing. Git and GitHub supported version control.

Category	Technology	Purpose
Core language	Python 3.11.9	Feature extraction, training, inference, and orchestration
Backend	FastAPI	Request handling and prediction endpoints
Interface	HTML, CSS, JavaScript, Bootstrap, Jinja	Multimodal acquisition and result presentation
Text representation	all-mpnet-base-v2	Sentence-level semantic embeddings
Audio representation	WavLM-Base-Plus and Librosa	Learned and engineered acoustic features
Vision representation	CLIP ViT-Large-Patch14	Image embeddings
Keystroke Module	Guided Record-Linkage Data Collector	Behavioural biometric feature acquisition
Machine Learning	Scikit-learn	Classification, pipelines, validation, and persistence
Data Processing	NumPy and Pandas	Feature engineering and schema management

Table 4.1. Principal implementation technologies

4.2.2 Reproducibility and Repository Management

The project was developed in Visual Studio Code, with Git and GitHub used for version control, incremental change tracking, and preservation of the implementation history. A Python virtual environment isolated project dependencies, while requirements.txt records the packages required to reconstruct the software environment.

The repository is organised into separate components for data collection, modality-specific feature extraction, classifier training, integrated inference, testing, and application deployment. Owing to repository storage constraints and the sensitive nature of behavioural data, the complete research dataset and large pretrained model weights are not distributed through GitHub. Instead, the repository contains a balanced demonstration dataset generated using “create_sample_dataset.py”, together with utilities for retrieving or restoring the required pretrained models.

This approach provides a lightweight, executable research artefact while distinguishing the distributable demonstration materials from the complete experimental dataset used during evaluation. Repository access instructions and a summary of the principal software artefacts are provided in Appendix A.

4.3 Front-End and Back-End Implementation

The interface provides text entry with concurrent keystroke recording, audio input, visual input, readiness feedback, and a prediction panel. Keystroke events are captured using JavaScript key-down and key-up listeners with high-resolution browser timestamps. Audio and image data are acquired through browser media APIs or file upload controls.

Client-side validation checks basic completeness and supported input types before submission. The interface then transmits the available data to FastAPI using asynchronous HTTP requests, preventing a full-page reload during inference.

The backend is implemented using FastAPI, which supports structured request handling, validation, file submission, and JSON responses (FastAPI, n.d.). Within SenseFuzeAI, the prediction workflow validates each request, invokes the appropriate modality processors, verifies the generated feature schemas, constructs the fused feature vector, applies the persisted classifier, and returns the predicted class, probabilities, confidence information, active modalities, and runtime diagnostics.

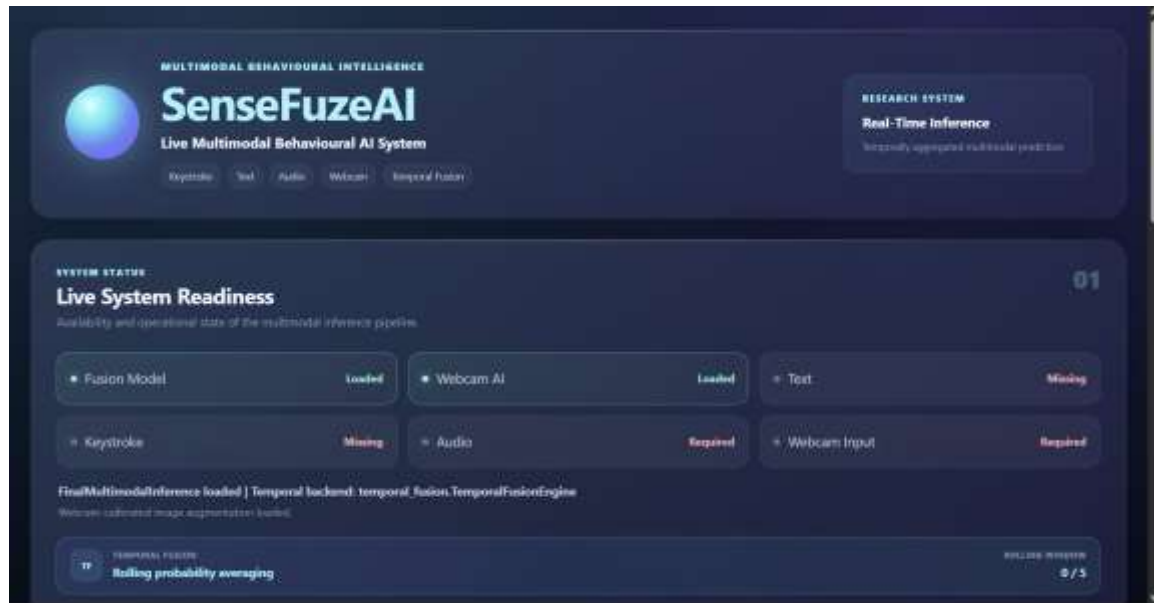
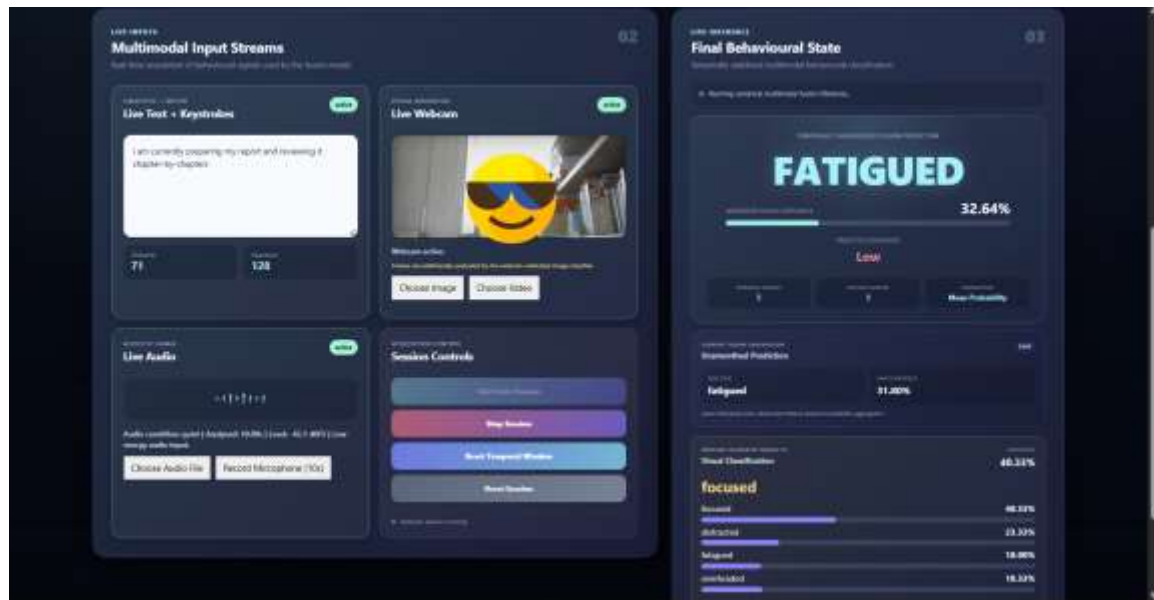
(a) System readiness and temporal-fusion status**(b) Active multimodal acquisition and session controls**

Figure 4.1. Implemented SenseFuzeAI integrated dashboard: (a) system-readiness and temporal-fusion status; and (b) active multimodal acquisition interface showing live text and keystroke capture, webcam input, audio processing, and session controls.

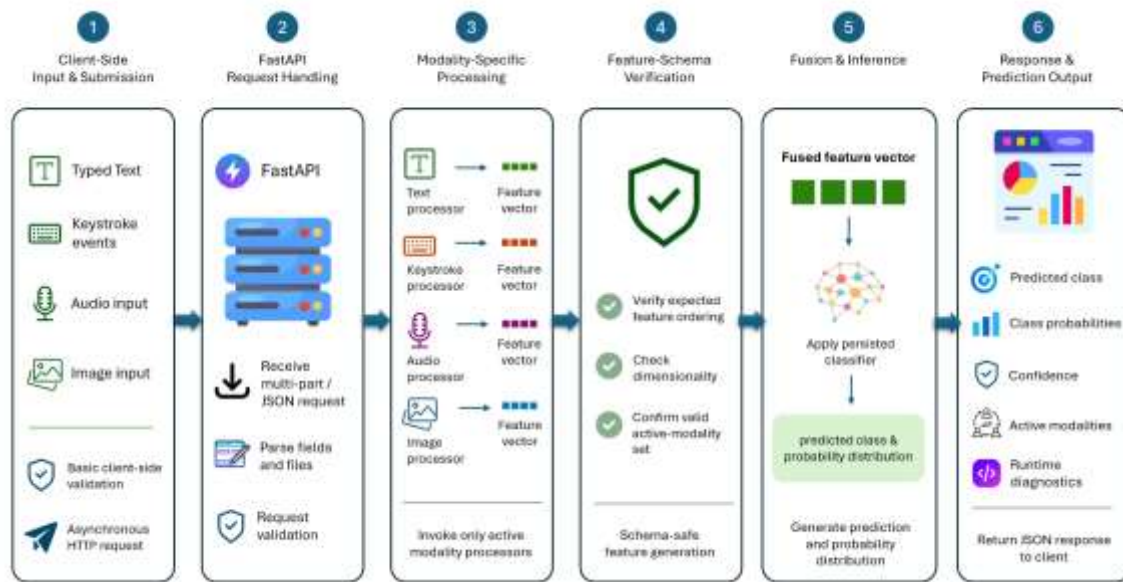


Figure 4.2. Back-end multimodal inference workflow.

4.4 Text and Keystroke Modules

4.4.1 Text processing

The text module converts each session's textual content into a normalised 768-dimensional sentence embedding using all-mpnet-base-v2 (Sentence Transformers, n.d.). The extraction script validates the session identifier, class label, and textual content before encoding the sample. Embeddings are stored with their session identifiers and labels so that they can later be aligned with the other modalities.

Feature extraction is separated from classifier training. This avoids repeatedly executing MPNet during model comparison and ensures that all candidate classifiers receive identical text representations.

The training pipeline evaluates conventional supervised classifiers using stratified five-fold cross-validation. Macro-F1 is used as the principal model-selection measure because it assigns equal importance to each behavioural class. The selected pipeline, ordered feature schema, and model metadata are persisted for deployment.

During application inference, new text is encoded through the same MPNet procedure and arranged according to the saved schema before classification. This ensures consistency between offline evaluation and online use.

4.4.2 Keystroke processing

The keystroke module analyses how text is entered rather than its semantic content. Raw key-down and key-up events are converted into features including typing duration, inter-key delay, hold time, typing speed, pause frequency, correction ratio, rhythm consistency, and burstiness.

Active-key tracking prevents repeated key-down events generated by a held key from being treated as separate physical presses. Event-pair validation identifies incomplete key-down and key-up sequences, while minimum-event checks reduce the risk of generating unstable statistics from very short samples.

A simplified extraction procedure is:

1. order the captured events by timestamp;
2. compute delays between consecutive key-down events;
3. match key-down and key-up events to estimate hold times;
4. calculate pause, correction, speed, rhythm, and variability features;
5. arrange the resulting values according to the stored schema.

Non-predictive columns, including labels, session identifiers, file paths, notes, and raw text, are explicitly excluded before training to reduce data leakage. The selected keystroke classifier and its feature schema are stored separately so that live events can be transformed into the identical feature order used during training.

The implementation also derives inspectable indicators describing rhythm stability, pausing, correction activity, and interaction speed. These indicators support explanation but do not independently determine the predicted class.

4.5 Audio and Vision Modules

The audio module combines pretrained and engineered features. Audio is converted to mono, resampled to 16 kHz, validated, and truncated to a bounded duration to control inference cost. WavLM-Base-Plus produces a fixed-length learned representation by pooling its temporal hidden states (Chen et al., 2022). Librosa supplements this representation with engineered descriptors including root-mean-square energy, zero-crossing rate, spectral centroid, bandwidth, roll-off, pitch statistics, and Mel-frequency cepstral coefficients (McFee et al., 2015).

The combined values are cleaned to remove non-finite outputs and stored with the corresponding session identifier and label. The selected audio classifier, schema, and metadata are subsequently persisted for deployment. The same feature-extraction procedure is applied to uploaded or recorded audio during inference.

The visual module uses CLIP ViT-Large-Patch14 as a general visual-semantic feature extractor (Radford et al., 2021). Each image is validated, converted to RGB, transformed through the CLIP processor, and encoded as a normalised image embedding. The resulting features are stored by session identifier and label before classifier comparison and model persistence.

To reduce the mismatch between the original online image data and live webcam input, an additional webcam-domain calibration pipeline was implemented. Representative frames were sampled from labelled webcam-style videos and encoded using the same frozen 768-dimensional CLIP representation used during live inference. The downstream image classifier was then retrained using these embeddings while preserving the pretrained CLIP encoder. Group-aware partitioning by source video prevented frames from the same video appearing in both training and evaluation subsets, reducing frame-level leakage.

CLIP embeddings are treated as general contextual representations rather than explicit measurements of gaze, posture, fatigue, or workload. Their behavioural value is therefore determined through the experimental comparisons reported in Chapter 5.

4.6 Multimodal Data Collection and Record Linkage

A dedicated collection application was implemented to support structured multimodal input and consistent record identification. Each application record receives a unique `session_id` linking its available text, keystroke events, audio file, image, operational label, validation status, and derived metadata. Within live application use, text and keyboard events can be captured concurrently, while audio and image inputs are validated and associated with the same application record where available.

The central registry provides deterministic joins between modality-specific feature tables and records modality availability, acquisition metadata, quality checks, and optional notes. The `session_id` therefore acts primarily as a record-linkage mechanism for schema construction and inference coordination.

The final 309-record evaluation dataset was not produced entirely through synchronous four-modality collection. Keystroke observations were collected from one participant, text samples were manually authored under class-guided conditions, and audio and visual samples were obtained from online sources and associated with the operational classes. Consequently, the evaluation dataset should be treated as a class-aligned composite dataset. Shared identifiers provide deterministic record linkage but do not imply common participant identity, simultaneous acquisition, or a naturally occurring shared behavioural episode.

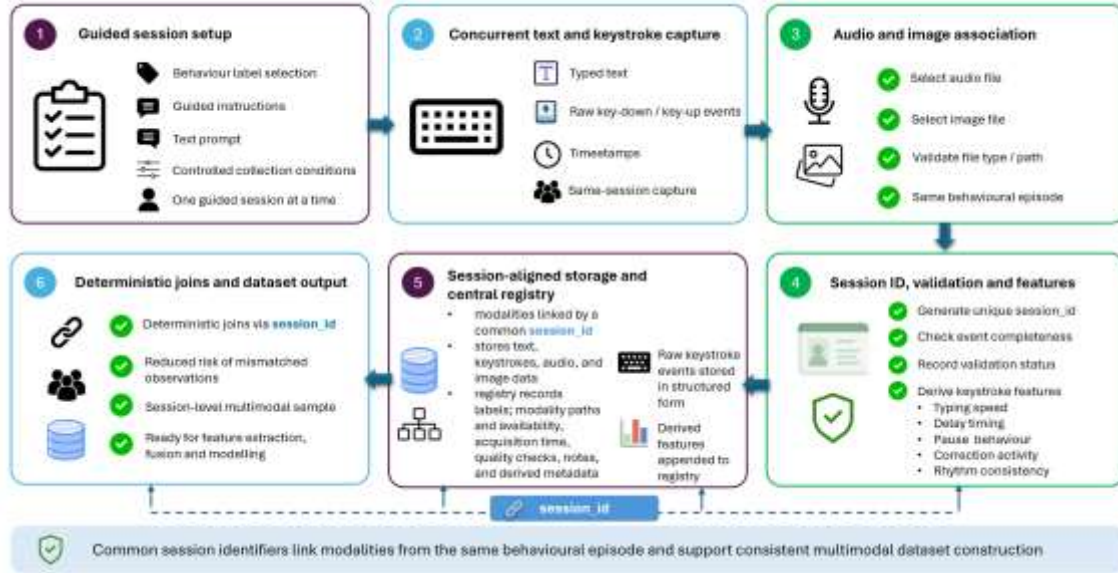


Figure 4.3. Guided multimodal data-collection and record-linkage interface.

4.7 Fusion Engine and Explainability

During offline training, the fusion engine loads the independently generated modality-specific feature tables and performs deterministic record-level joins using the shared session_id together with the operational class label. This procedure constructs schema-consistent composite records but does not imply temporal synchronisation of the underlying source observations. Invalid, incomplete, or non-conforming records are excluded before fusion, while identifiers, labels, file paths, and other non-predictive metadata are removed from the feature matrix.

For each retained composite record, the implementation first concatenates the four principal modality representations:

$$\mathbf{x}_M^{(i)} = \mathbf{x}_K^{(i)} \parallel \mathbf{x}_T^{(i)} \parallel \mathbf{x}_A^{(i)} \parallel \mathbf{x}_V^{(i)} \in R^{2367}$$

Here, $\mathbf{x}_M^{(i)}$ denotes the principal four-modality representation, comprising 22 engineered keystroke features, a 768-dimensional MPNet textual embedding, an 809-dimensional WavLM-Librosa audio representation, and a 768-dimensional CLIP visual embedding. The six additional derived predictors retained in the persisted schema are represented by and are appended to form the complete classifier input:

$$\mathbf{x}_F^{(i)} = \mathbf{x}_M^{(i)} || \mathbf{x}_D^{(i)} \in R^{2373}$$

The implementation preserves the complete ordered 2,373-feature schema so that the same classifier input can be reconstructed consistently during deployment.

The fused vector is supplied to the trained classifier:

$$\hat{b} = g_{\theta}(x_F)$$

where $\hat{b}$ is one of the four operational behavioural labels: focused, distracted, fatigued, or overloaded.

Candidate fusion classifiers are compared using stratified five-fold cross-validation. Accuracy and macro-F1 are recorded, with macro-F1 used as the principal model-selection criterion. The selected pipeline is then retrained using the available fused observations and serialised together with its preprocessing components, ordered feature schema, class definitions, and model metadata.

During application inference, the system does not use a behavioural label or join separate tables. Instead, each modality processor generates its feature vector from the current input sample, after which the vectors are validated and arranged according to the stored schema before classification.

For each prediction, the classifier returns a probability distribution over the four classes. The class assigned the highest probability is reported as the predicted state. The implementation also calculates the difference between the two largest probabilities as a confidence gap. This value indicates how decisively the classifier separates its two leading alternatives, but it should not be interpreted as a calibrated probability that the inferred state is psychologically correct.

Explainability is supported through the displayed class distribution, confidence gap, active modality information, runtime diagnostics, and selected keystroke indicators. These outputs permit inspection of the computational evidence associated with a prediction, but they do not constitute a causal explanation of the user's internal state.

4.8 Integrated Web Application

The final application integrates acquisition, modality-specific feature extraction, multimodal fusion, classification, and presentation within a single browser-based workflow. Text and keystroke events are captured directly through the interface, while audio and visual inputs are acquired through browser microphone and webcam access subject to device and browser support.

Before fused inference, the backend validates the required inputs and reconstructs the complete **2,373-dimensional feature vector** according to the persisted fusion schema. Schema reconstruction enforces the expected feature ordering and dimensionality before classification, while the response records modality and prediction information for presentation and logging. Classifier sensitivity to unavailable modality information is evaluated separately in Section 5.6.2 using controlled schema-preserving zero masking. This experiment is a synthetic inference-time stress test and does not imply that every physically absent-modality condition is natively supported by the deployed application.

Prediction requests are submitted asynchronously from the browser, allowing the interface to remain responsive while the backend performs modality-specific feature extraction, schema reconstruction, and fusion inference. Computational responsiveness is evaluated through the repeated CPU-based latency benchmark reported in Section 5.7, which separates cold initialisation, modality-specific processing, fusion classification, and warm end-to-end inference from acquisition delays such as typing, microphone recording, and temporal waiting. Prediction results are displayed dynamically and recorded in structured logs containing timestamps, predicted states, class probabilities, confidence gaps, feature dimensionality, and modality-availability information.

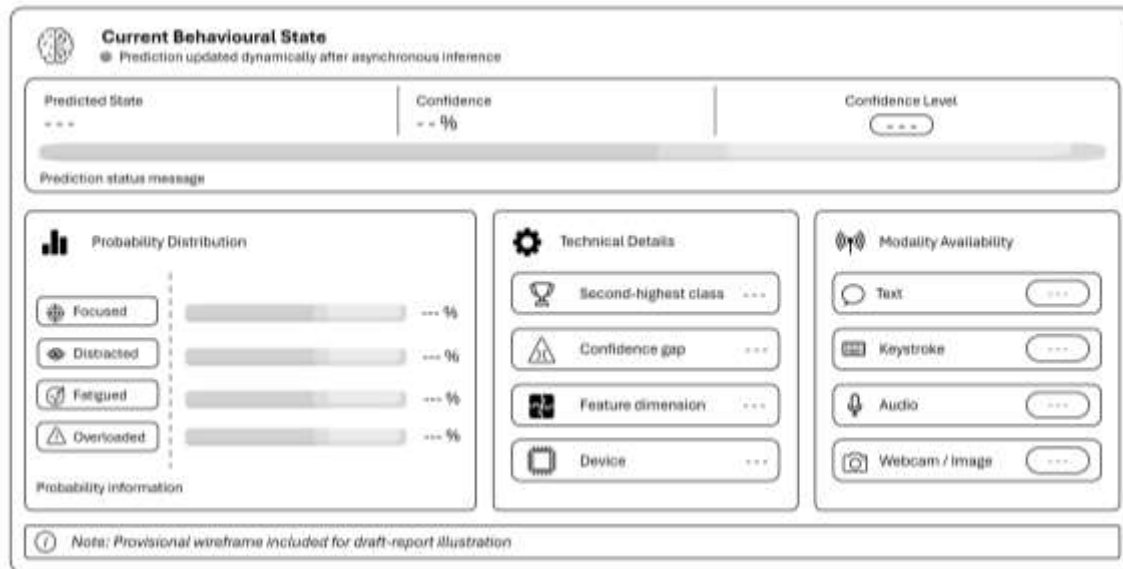


Figure 4.4. Multimodal prediction and probability display.

4.9 Key Challenges and Resolutions

Three implementation challenges were particularly significant.

First, independently constructed modality samples could not be assumed to be temporally synchronised observations of the same naturally occurring behavioural episode. Shared session identifiers and a central metadata registry were therefore used to provide deterministic record-level association and consistent schema joins. This resolved the engineering problem of record linkage, but not the methodological limitation of non-contemporaneous collection.

Second, high-dimensional features created a substantial risk of mismatch between training and deployment. Ordered feature schemas, explicit missing-column checks, and persisted preprocessing pipelines were therefore introduced to preserve feature ordering and dimensional consistency.

Third, pretrained-model inference introduced substantially greater computational cost than conventional engineered-feature calculations. Offline feature caching, bounded audio duration, reusable model instances, and asynchronous request handling were incorporated to reduce avoidable processing overhead. Their computational effect was quantified through the repeated CPU-based latency evaluation reported in Section 5.7, which separated cold initialisation, modality-specific processing, fusion classification, and warm end-to-end inference.

Additional safeguards included keystroke-event validation, exclusion of metadata from training, structured exception handling, prediction logging, and automated unit, integration, system, and acceptance tests.

These measures transformed the initial collection of modality-specific prototypes into a reproducible integrated application. Chapter 5 evaluates predictive performance, modality contribution, classifier sensitivity under simulated modality loss, computational responsiveness, operational behaviour, and the verified software-testing evidence produced by the implemented system.

CHAPTER 5: EVALUATION (2446/2500 words)

5.1 Evaluation Objectives and Research-Question Mapping

SenseFuzeAI was evaluated at two distinct levels: **predictive discrimination** and **integrated-system performance**. Predictive evaluation examined unimodal and fused classification, modality contribution, and missing-modality sensitivity; system evaluation examined schema consistency, software verification, interpretability, and processing latency. These evidence types are not interchangeable: successful execution does not establish behavioural validity, and high classification performance does not establish participant-independent recognition.

Stratified five-fold cross-validation, unimodal baselines, retrained ablation, inference-time zero masking, latency benchmarking, and automated testing addressed complementary aspects of RQ1-RQ4.

Because the records are class-aligned composites rather than naturally co-recorded behavioural episodes, and the principal keystroke observations originate from one participant, the results establish **within-dataset proof-of-concept feasibility** rather than population-level behavioural-state validity.

Research question	Evaluation focus	Experimental evidence	Principal measures
RQ1: What software architecture can coordinate modality-specific models and record-aligned features to support near-real-time behavioural inference?	Architectural integration, operational execution, and computational responsiveness	Record linkage; feature-schema validation; modality-component integration; web-based inference; persisted pipeline execution; automated software testing; repeated cold-start and warm inference benchmarking; modality-specific	Successful pipeline execution; schema consistency; verified endpoints; application behaviour; automated-test outcomes; mean/median latency; latency standard deviation; p95 latency; cold-start and warm end-to-end processing time

		feature-extraction timing; end-to-end processing-latency measurement	
RQ2: How does multimodal fusion compare with individual modalities in predictive performance and robustness?	Unimodal versus multimodal predictive performance; modality contribution; predictive robustness under modality loss	Five-fold cross-validated unimodal and complete-fusion comparison; supplementary confusion-matrix analysis; leave-one-modality-out ablation; inference-time missing-modality stress testing; classifier-level schema-preserving zero-masking stress testing	Accuracy; macro-F1; per-class precision/recall/F1; confusion matrices; absolute fusion improvement; macro-F1 under modality ablation; macro-F1 under missing-modality conditions
RQ3: How can low-level keystroke features be transformed into interpretable behavioural indicators, and what contribution do these indicators make to classification?	Keystroke abstraction, interpretability, standalone predictive value, and incremental contribution within multimodal fusion	Keystroke-only classifier evaluation; inspection of engineered timing, pausing, correction, rhythm, speed, and variability indicators; leave-keystroke-out fusion ablation using the common validation protocol	Keystroke accuracy and macro-F1; class-level errors; qualitative interpretability of behavioural indicators; macro-F1 without keystrokes; macro-F1 relative to complete fusion
RQ4: What trade-offs between	Whole-system trade-off analysis	Comparative unimodal and	Predictive metrics; 0.0097 absolute

predictive performance, computational responsiveness, robustness, and interpretability are observed in the implemented prototype?	across predictive quality, dimensionality, latency, robustness, input requirements, and interpretability	fusion results; 2,373-dimensional complete classifier input; leave-one-modality-out ablation; missing-modality stress testing; repeated latency benchmarking; software robustness testing; modality requirements; interface-level interpretability	macro-F1 fusion improvement; feature dimensionality; modality-ablation degradation; missing-modality degradation; cold-start and warm latency; p95 latency; input requirements; interpretability observations
---	--	--	---

Table 5.1. Mapping between research questions, evaluation focus, experimental evidence, and principal measures

5.2 Dataset and Experimental Protocol

5.2.1 Experimental Environment and Reproducibility

Experiments were executed in Python 3.11.9 on CPU-only hardware. MPNet, WavLM, and CLIP were used as fixed pretrained feature extractors rather than fine-tuned on the 309-record dataset. Reproducibility controls included fixed random seeds, persisted preprocessing and classification pipelines, explicit feature schemas, version-controlled evaluation scripts, and fold-local preprocessing so that validation data did not influence fitted transformations. The same CPU environment was used for the repeated latency evaluation in Section 5.7.

Component	Configuration
Execution device	CPU only
GPU acceleration	Not used
Python version	Python 3.11.9
Machine-learning framework	scikit-learn
Deep-learning framework	PyTorch
Pretrained-model libraries	SentenceTransformers; Hugging Face Transformers
Audio-processing library	Librosa
Deployment framework	FastAPI; Jinja2
Reproducibility controls	Fixed random seed; persisted pipelines; explicit feature schemas; version-controlled scripts

Table 5.2. Experimental software environment

5.2.2 Dataset Construction, Provenance, and Composition

The principal dataset contained 309 complete class-aligned composite records: 77 Focused, 77 Distracted, 77 Fatigued, and 78 Overloaded. Each record carried a unique session_id, enabling deterministic linkage between modality-specific feature tables. This identifier does not imply simultaneous acquisition, common participant identity, or observation of one naturally occurring behavioural episode.

Text was manually authored under class-guided conditions, keystrokes originated from one participant, and audio and visual observations were obtained from heterogeneous external sources. These characteristics make the dataset suitable for controlled proof-of-concept comparison of representations, while limiting ecological, participant-independent, and construct validity.

Behavioural state	Recorded sessions	Keystroke records retained	Text records retained	Audio records retained	Visual records retained	Complete fused records
Focused	77	77	77	77	77	77
Distracted	77	77	77	77	77	77
Fatigued	77	77	77	77	77	77
Overloaded	78	78	78	78	78	78
Total	309	309	309	309	309	309

Table 5.3. Composition of the class-aligned experimental dataset

The 20-record demonstration dataset was excluded from predictive evaluation. The four labels are operational experimental categories rather than clinically validated psychological states; evaluation therefore concerns discrimination among project-defined classes.

5.2.3 Feature Representations and Multimodal Construction

Each composite record contained 22 engineered keystroke features, a 768-dimensional all-mpnet-base-v2 text embedding, 809 audio features comprising 768 WavLM dimensions and 41 Librosa descriptors, and a 768-dimensional CLIP visual embedding. Six additional derived predictors retained in the persisted schema produced the final 2,373-dimensional classifier input.

Feature group	Dimensionality
Keystroke features	22
Text embedding	768
Audio representation	809
Visual embedding	768
Additional derived predictors retained in final schema	6
Final fused representation	2,373

Table 5.4. Feature dimensionality of the final multimodal representation

For record i , the complete fused representation is

$$\mathbf{x}_F^{(i)} = [\mathbf{x}_K^{(i)}, \mathbf{x}_T^{(i)}, \mathbf{x}_A^{(i)}, \mathbf{x}_V^{(i)}, \mathbf{x}_D^{(i)}], \quad \mathbf{x}_F^{(i)} \in R^{2373},$$

where $\mathbf{x}_D^{(i)} \in \mathbb{R}^6$ denotes the six additional derived predictors retained in the final schema.

The 2,373-predictor, 309-record setting creates a pronounced sample-to-dimensionality imbalance and therefore increases overfitting risk. Although fixed pretrained extractors, leakage-safe preprocessing, and fold-local fitting reduce avoidable methodological error, they do not eliminate the possibility that high-dimensional representations capture source-specific or class-construction regularities. This is especially relevant when interpreting perfect cross-validation performance, which may reflect genuine complementarity, redundant class-associated cues, or both. The resulting performance is therefore interpreted cautiously as within-dataset evidence rather than external validation.

5.2.4 Training and validation protocol

Model comparison used common stratified five-fold cross-validation, with preprocessing transformations fitted only within each training fold. Candidate classifiers were manually specified rather than selected through unrestricted automated search. A stratified 80:20 split provided supplementary confusion-matrix and class-level inspection.

Because this supplementary partition participated in model comparison rather than remaining completely untouched after all analytical decisions, it is treated as supporting evidence rather than an independent estimate of external generalisation. The protocol therefore supports controlled within-dataset comparison, not unseen-user or unseen-domain validation. This distinction is important because repeated model selection and evaluation on limited data can produce optimistic estimates (Cawley and Talbot, 2010).

Parameter	Configuration
Primary evaluation dataset	309 near-balanced composite records
Number of classes	4
Class distribution	77 focused, 77 distracted, 77 fatigued, and 78 overloaded
Final predictive dimensionality	2,373 features
Primary validation method	Stratified five-fold cross-validation
Fold shuffling	Enabled
Random seed	42
Primary selection metric	Mean macro-F1
Complementary metric	Mean accuracy
Variability measure	Standard deviation across five folds
Core candidate classifiers	Logistic Regression; Random Forest; SVM-RBF
Additional classifiers	XGBoost, LightGBM and CatBoost, where executed
Hyperparameter procedure	Fixed manually specified configurations
Feature scaling	Applied within each scikit-learn pipeline
Supplementary evaluation	Stratified 80:20 split
Supplementary outputs	Per-class classification report; confusion matrix
Generalisation scope	Within-dataset proof-of-concept; keystroke evidence participant-specific

Table 5.5. Training, validation, and model-comparison protocol

5.3 Evaluation Metrics

Accuracy summarised overall classification correctness, while macro-precision, macro-recall, and macro-F1 gave equal contribution to each operational class. Macro-F1 was treated as the principal comparative metric because aggregate accuracy can obscure uneven class performance (Sokolova and Lapalme, 2009). Confusion matrices were additionally inspected to identify systematic class-level errors.

Cross-validation results are reported as the mean and population standard deviation across five folds. Absolute differences in macro-F1, denoted ΔF_1 , are used where they clarify the incremental effect of fusion or modality removal. These values describe observed differences within the dataset and are not inferential evidence of population-level superiority.

The application also reports a confidence gap,

$$g = p_1 - p_2,$$

where p_1 and p_2 are the two largest class probabilities. Because probability calibration was not independently evaluated, the gap is interpreted as relative separation between model outputs, not as a calibrated probability that the prediction is correct.

5.4 Individual Modality Results

5.4.1 Text Model

The 768-dimensional MPNet representation was highly discriminative within the constructed dataset. Logistic Regression achieved mean cross-validation macro-F1 of **0.9871**, while the supplementary split achieved **0.9839**, with one Overloaded observation classified as Distracted.

The result demonstrates strong semantic separability of the authored class-aligned text samples. It does not establish unconstrained behavioural-state recognition because class-guided writing may introduce lexical or semantic regularities correlated with the experimental labels.

Classifier	Mean accuracy	Accuracy SD	Mean macro-F1	Macro-F1 SD
Logistic Regression	0.9871	0.0121	0.9871	0.0121
SVM-RBF	0.9839	0.0177	0.9839	0.0177
CatBoost	0.9741	0.0078	0.9740	0.0078
LightGBM	0.9644	0.0158	0.9645	0.0158
XGBoost	0.9612	0.0241	0.9613	0.0240
Random Forest	0.9612	0.0241	0.9608	0.0240

Table 5.6. Five-fold cross-validation results for the MPNet text representation

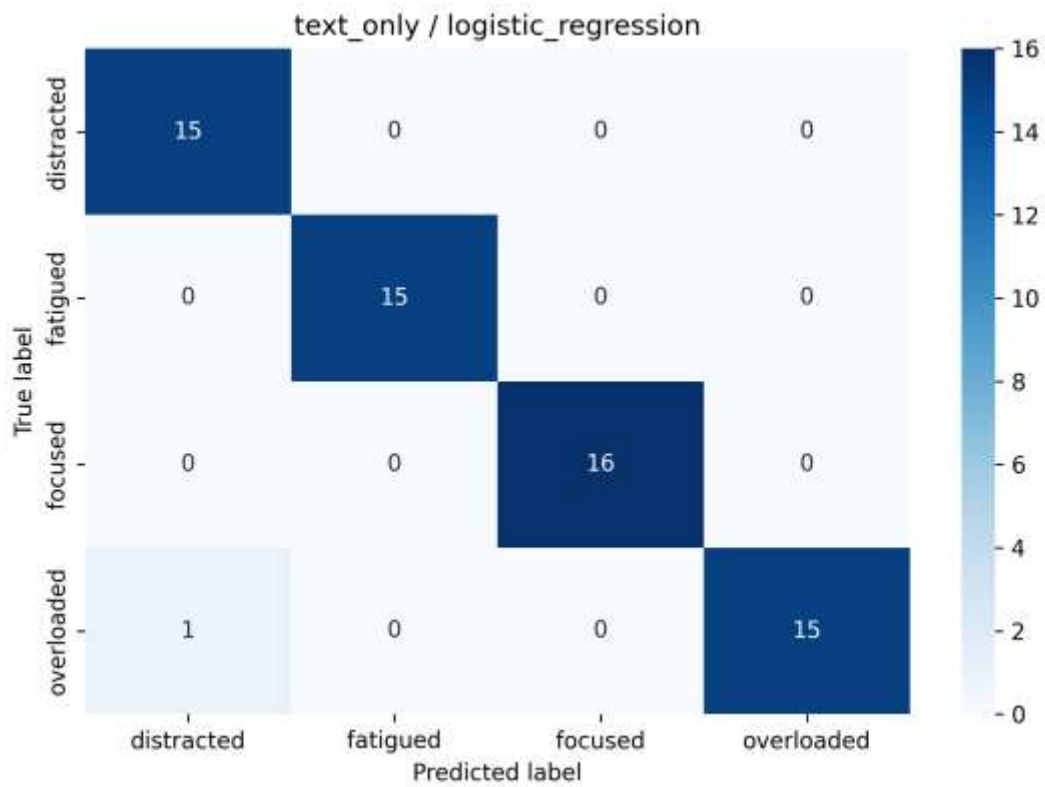


Figure 5.1. Confusion matrix for the selected MP-Net-Logistic Regression text classifier on the supplementary test split.

5.4.2 Keystroke Model

The 22-feature engineered keystroke representation produced the strongest unimodal cross-validation result. Random Forest achieved mean macro-F1 of **0.9903**, while the supplementary split achieved **0.9838**, with one Distracted observation classified as Overloaded.

This is notable because a compact engineered representation nearly matched complete multimodal performance despite containing far fewer predictors than the pretrained text, audio, and visual spaces. Timing, pausing, correction, rhythm, speed, and variability therefore contain substantial discriminatory information within the recorded conditions.

However, the keystroke data originate from one participant. The experiment cannot determine whether the observed separation reflects general behavioural variation, participant-specific typing habits, task structure, or interactions among these factors. The result therefore supports RQ3 as interpretable within-participant discrimination, not participant-independent behavioural measurement.

Classifier	Mean accuracy	Accuracy SD	Mean macro-F1	Macro-F1 SD
Random Forest	0.9903	0.0129	0.9903	0.0130
Logistic Regression	0.9871	0.0065	0.9872	0.0064
LightGBM	0.9871	0.0188	0.9871	0.0187
SVM-RBF	0.9871	0.0121	0.9870	0.0122
CatBoost	0.9839	0.0144	0.9838	0.0145
XGBoost	0.9838	0.0102	0.9837	0.0103

Table 5.7. Five-fold cross-validation results for the keystroke representation

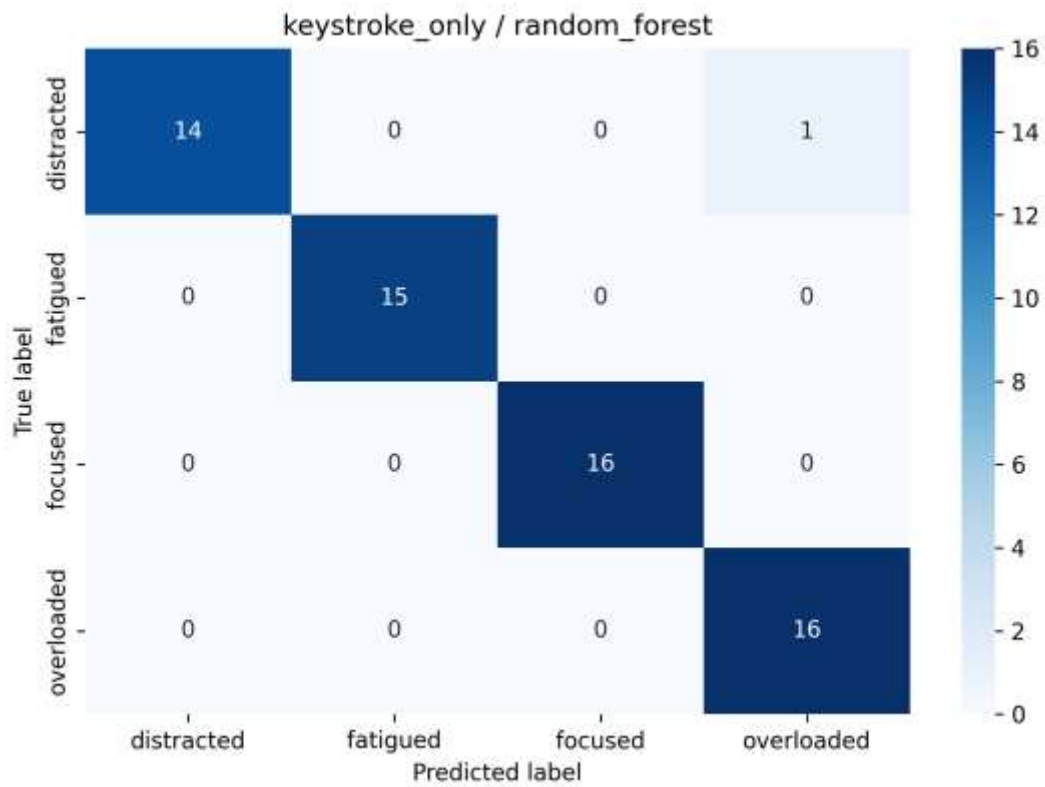


Figure 5.2. Confusion matrix for the selected keystroke-only Random Forest classifier on the supplementary stratified test split.

5.4.3 Audio Model

The WavLM-Librosa representation was the weakest individual modality. SVM-RBF achieved mean cross-validation macro-F1 of **0.8496** and supplementary macro-F1 of **0.8678**, with the principal confusion occurring between Fatigued and Overloaded.

This weaker result remains informative. It suggests that the available acoustic representation provides less stable class separation than text, keystrokes, or vision. Heterogeneous speaker and recording provenance prevents attributing these errors solely to behavioural ambiguity.

Classifier	Mean accuracy	Accuracy SD	Mean macro-F1	Macro-F1 SD
SVM-RBF	0.8511	0.0427	0.8496	0.0442
LightGBM	0.8444	0.0530	0.8424	0.0545
Random Forest	0.8350	0.0373	0.8355	0.0379
XGBoost	0.8249	0.0630	0.8251	0.0624
Logistic Regression	0.8122	0.0341	0.8096	0.0364
CatBoost	0.7959	0.0602	0.7933	0.0629

Table 5.8. Five-fold cross-validation results for the combined WavLM-Librosa audio representation

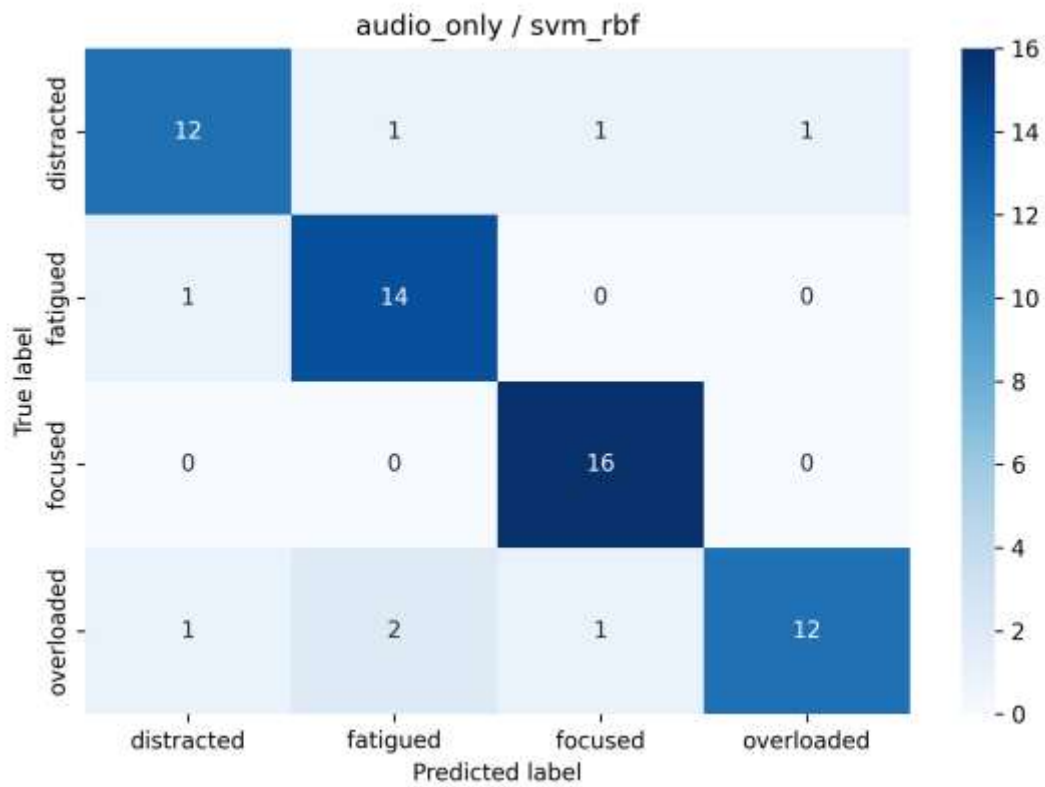


Figure 5.3. Confusion matrix for the selected combined WavLM-Librosa SVM-RBF audio classifier on the supplementary stratified test split.

5.4.4 Image Model

The CLIP visual representation achieved mean cross-validation macro-F1 of **0.9351** using SVM-RBF and supplementary macro-F1 of **0.9514**. Fatigued observations were most frequently confused with Overloaded observations.

Because CLIP is a transferable visual-semantic representation rather than a specialised detector of gaze, fatigue, or workload, its embeddings may encode background, scene, clothing, pose, and other contextual regularities alongside behaviourally relevant cues. The result therefore indicates visual-semantic separability, not validated behavioural-state detection.

Across the unimodal models, text and keystrokes were almost perfectly separable, whereas audio and vision showed greater Fatigued-Overloaded confusion. This pattern may indicate overlapping observable cues, but heterogeneous source provenance prevents attributing the confusion solely to behavioural similarity.

Classifier	Mean accuracy	Accuracy SD	Mean macro-F1	Macro-F1 SD
SVM-RBF	0.9354	0.0367	0.9351	0.0370
Logistic Regression	0.9160	0.0372	0.9157	0.0376
Random Forest	0.9161	0.0503	0.9153	0.0506
CatBoost	0.9096	0.0484	0.9093	0.0489
LightGBM	0.9032	0.0603	0.9035	0.0600
XGBoost	0.8838	0.0622	0.8837	0.0622

Table 5.9. Five-fold cross-validation results for the CLIP visual representation

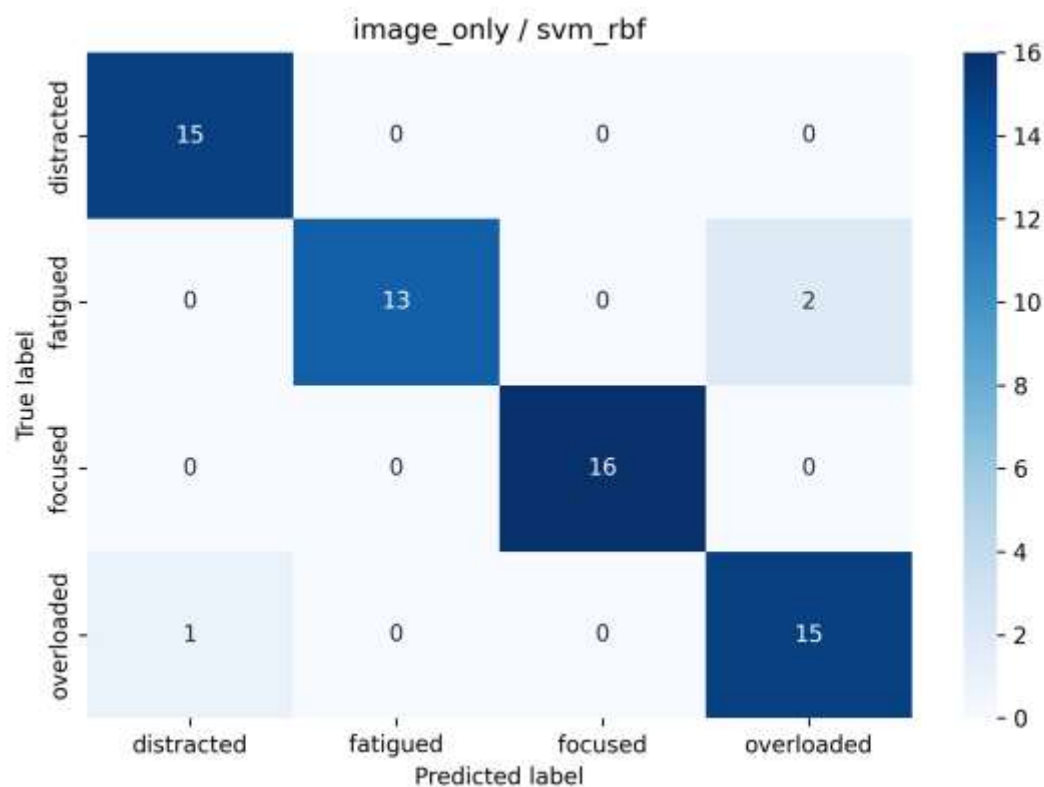


Figure 5.4. Confusion matrix for the selected CLIP-SVM-RBF image classifier on the supplementary stratified test split.

5.5 Unimodal and Multimodal Classification Performance

The unimodal comparison shows that dimensionality alone did not determine predictive performance: the 22-feature keystroke representation outperformed the substantially larger audio and visual representations. This motivates evaluating complete fusion in terms of incremental value rather than raw dimensionality.

The fused representation contained 2,373 predictors. Logistic Regression and Random Forest both achieved mean five-fold accuracy and macro-F1 of **1.0000**, while the supplementary split was also classified without error. Random Forest was retained for deployment, although the evaluation provides no evidence that it was predictively superior to Logistic Regression.

Classifier	Mean accuracy	Accuracy SD	Mean macro-F1	Macro-F1 SD
Logistic Regression	1.0000	0.0000	1.0000	0.0000
Random Forest	1.0000	0.0000	1.0000	0.0000
XGBoost	0.9968	0.0065	0.9968	0.0065
CatBoost	0.9968	0.0065	0.9968	0.0065
LightGBM	0.9935	0.0129	0.9936	0.0127
SVM-RBF	0.9935	0.0080	0.9935	0.0079

Table 5.10. Five-fold cross-validation results for the complete multimodal representation

Representation	Selected classifier	Accuracy	Macro-precision	Macro-recall	Macro-F1
Keystroke	Random Forest	0.9839	0.9853	0.9833	0.9838
Text	Logistic Regression	0.9839	0.9844	0.9844	0.9839
Audio	SVM-RBF	0.8710	0.8732	0.8708	0.8678
Vision	SVM-RBF	0.9516	0.9550	0.9510	0.9514
Complete fusion	Random Forest	1.0000	1.0000	1.0000	1.0000

Table 5.11. Supplementary test-set comparison of selected unimodal and complete-fusion models

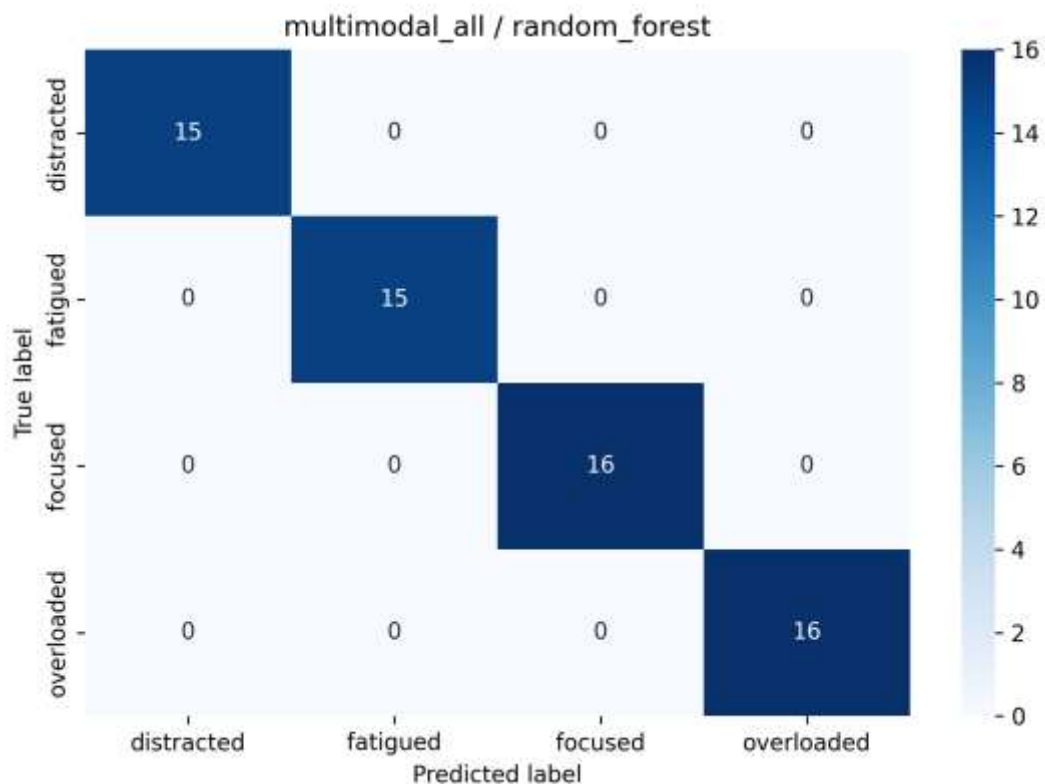


Figure 5.5. Confusion matrix for the selected complete-fusion Random Forest classifier on the supplementary stratified test split.

Complete fusion therefore achieved the strongest observed within-dataset performance, but its incremental advantage was small. Relative to keystroke Random Forest ($F_1 = 0.9903$), fusion improved mean macro-F1 by only **0.0097**. This gain must be weighed against the increase from 22 to 2,373 predictors and the requirement to execute several pretrained feature extractors.

The perfect score also creates a **ceiling effect**: once performance reaches 1.0000, additional unique contribution cannot appear as further accuracy. Class-guided text, participant-specific keystrokes, and heterogeneous audiovisual sources may also introduce class-associated regularities that are easier to discriminate than naturally occurring behavioural states. The fused result therefore establishes a highly discriminative within-dataset representation, not universal behavioural-state recognition.

5.6 Modality Contribution and Missing-Modality Robustness

5.6.1 Leave-One-Modality-Out Ablation

Leave-one-modality-out ablation tested whether remaining representations could compensate after retraining. For each condition, one modality group was removed and a new Random Forest was fitted under the common five-fold protocol.

Removing text produced the largest decrease, from macro-F1 **1.0000 to 0.9935** ($\Delta F_1 = 0.0065$), while removing vision reduced macro-F1 to **0.9968** ($\Delta F_1 = 0.0032$). Removing either keystrokes or audio produced no measurable decrease.

Although keystrokes were the strongest standalone modality, removing them did not reduce fused performance after retraining. Strong standalone predictive value therefore did not imply unique incremental information within this dataset. Because complete fusion is already saturated, however, near-zero degradation should not be interpreted as proof that a removed modality is intrinsically uninformative.

Configuration	Dimensions	Mean Accuracy	Accuracy SD	Mean Macro-F1	Macro-F1 SD	Δ Macro-F1
Full fusion	2,373	1.0000	0.0000	1.0000	0.0000	0.0000
Fusion-Keystroke	2,351	1.0000	0.0000	1.0000	0.0000	0.0000
Fusion-Text	1,605	0.9935	0.0129	0.9935	0.0130	0.0065
Fusion-Audio	1,564	1.0000	0.0000	1.0000	0.0000	0.0000
Fusion-Vision	1,599	0.9968	0.0065	0.9968	0.0065	0.0032

Table 5.12. Leave-one-modality-out ablation of the complete fusion classifier using the common stratified five-fold validation protocol

5.6.2 Missing-Modality Robustness

Inference-time zero masking tested a different property: sensitivity of the already-trained complete classifier when one modality was synthetically removed without retraining.

Vision masking produced the largest reduction, lowering macro-F1 from **1.0000 to 0.9190** ($\Delta F_1 = 0.0810$); keystroke masking reduced it to **0.9737** ($\Delta F_1 = 0.0263$), while text and audio masking caused no measurable reduction.

The contrast demonstrates that adaptability after retraining and robustness of a fixed deployed classifier are not equivalent. The model had learned some dependence on vision and keystroke features even though retraining could compensate for their removal.

These values nevertheless describe a schema-preserving synthetic missingness condition. Zero vectors may themselves be outside the original training distribution, so the experiment does not establish robustness to every physically absent webcam, microphone, or interaction stream.

Inference condition	Mean Accuracy	Mean Macro-F1	Macro-F1 SD	Δ Macro-F1
All modalities available	1.0000	1.0000	0.0000	0.0000
Keystroke unavailable	0.9741	0.9737	0.0170	0.0263
Text unavailable	1.0000	1.0000	0.0000	0.0000
Audio unavailable	1.0000	1.0000	0.0000	0.0000
Vision unavailable	0.9256	0.9190	0.0664	0.0810

Table 5.13. Predictive sensitivity of the complete fusion classifier under simulated single-modality zero masking

5.7 Near-Real-Time System and Latency Evaluation

Repeated CPU benchmarking separated cold initialisation from warm inference. Cold initialisation averaged **8,388.62 ms**. Across 30 warm repetitions, warm end-to-end processing averaged **2,501.15 ms**, with median **2,174.85 ms** and p95 **3,830.72 ms**. Component-level timings are reported in Table 5.13.

The mean supports interactive processing of bounded multimodal samples on the evaluated hardware, but the p95 exceeds the nominal 2.5-second operational update cadence. SenseFuzeAI is therefore characterised as a near-real-time interactive prototype, not a system providing a deterministic real-time deadline.

The latency profile also exposes the RQ4 trade-off: complete fusion gained only **0.0097 macro-F1** over the 22-feature keystroke model while requiring 2,373 predictors and multiple pretrained extractors. Predictive gain must therefore be weighed against computational cost and input requirements.

(a) **Temporally aggregated multimodal behavioural-state prediction**



(b) Raw fusion observation and webcam-calibrated modality evidence

Figure 5.6. SenseFuzeAI integrated prediction interface following a completed multimodal inference session: (a) temporally aggregated behavioural-state prediction with confidence and rolling-window status; and (b) the current raw fusion observation and webcam-calibrated visual-modality probability distribution.



Figure 5.7. SenseFuzeAI technical details interface showing classifier metadata, active modalities, feature dimensionality, temporal sample status, and both temporally aggregated and raw fusion probability distributions.

Component	n	Mean (ms)	SD (ms)	Median (ms)	p95 (ms)
Cold model/application initialisation	5	8388.62	1259.28	8209.43	10258.99
Keystroke processing	30	0.51	0.30	0.39	1.06
Text feature extraction	30	41.59	9.45	37.21	60.51
Audio feature extraction	30	906.67	809.16	710.83	2191.57
Visual feature extraction	30	1266.30	188.75	1256.16	1649.12
Fusion/classifier	30	134.63	29.42	128.90	194.57
Warm end-to-end processing	30	2501.15	824.60	2174.85	3830.72

Table 5.14. Repeated CPU-based latency measurements for the SenseFuzeAI inference pipeline

5.8 Software Verification

Automated verification provided complementary evidence concerning implementation correctness. The final pytest execution completed **158 tests successfully**, while the canonical project runner additionally exercised syntax, temporal-fusion mathematics, feature-schema contracts, browser-audio behaviour, dataset-comparison contracts, and evaluation assumptions. The single reported skip corresponded to an optional heavyweight model-initialisation check rather than a failed mandatory test.

The project-wide checks and pytest cases should not be added together as independent tests because the canonical runner incorporates the pytest execution as one verification stage.

This evidence supports repeatability of schema handling, temporal aggregation, request processing, and integration behaviour. It does **not** establish behavioural construct validity, participant-independent predictive generalisation, or reliable operation across arbitrary deployment hardware. Separating implementation correctness from empirical validity is therefore central to the project critique.

Test level	Scope	Actual outcome	Status
Unit	Probability normalisation, confidence logic, temporal aggregation, keystroke handling, malformed inputs, and backend utility behaviour	Included in the 158-case pytest suite; probability, confidence, keystroke, temporal-window, validation, and related utility contracts passed.	Pass
Integration	Model artefacts, original and webcam-calibrated visual models, calibration metadata, feature schemas, multimodal	Required artefacts and schema compatibility were verified, including the complete 2,373-feature fusion representation and multimodal	Pass

	inference modules, and shared temporal contracts	inference integration.	
System	FastAPI health/model-status endpoints, prediction gating, continuous microphone handling, session state, browser controls, and temporal behaviour	Endpoint responses, HTTP 409 input-gating behaviour, streaming-audio lifecycle, session isolation, and browser-facing temporal behaviour were verified successfully.	Pass
Acceptance	Required interfaces, MPNet, WavLM and CLIP artefacts, fusion schema, four-class output, webcam calibration, temporal architecture, and evaluation components	Required final-project implementation and evaluation artefacts were present and their principal architectural contracts were verified.	Pass
Dataset-comparison verification	EmoSurv-SenseFuzeAI 23-feature harmonisation, provenance, label contracts, identifiers, frozen train/test splits, participant/session leakage prevention, matched hold-outs,	Harmonised dataset and comparison-pipeline contracts were verified successfully without retraining the comparison models.	Pass

	and A-F experiment outputs		
Smoke / Regression	Critical files, temporal fusion, schema compatibility, web status, continuous-audio lifecycle, and final prediction behaviour	Critical cross-component functionality remained operational following implementation changes.	Pass

Table 5.15. Verified software-testing results for the final implementation

5.9 Critical Discussion Against Research Questions and Objectives

Interpretability in SenseFuzeAI is operational rather than causal. The interface exposes class probabilities, confidence gap, active-modality status, runtime diagnostics, and selected keystroke indicators. These support inspection of computational evidence but do not explain latent psychological state or establish causal relationships.

Taken together, the experiments answer the research questions at proof-of-concept level. **RQ1** is supported by successful record-linked orchestration, persisted schema enforcement, automated verification, and measured interactive latency. **RQ2** is supported by unimodal/fusion comparison and complementary modality-removal experiments: fusion maximised within-dataset performance, but robustness depended on both modality and evaluation condition.

RQ3 is supported by the strong 22-feature keystroke model and its interpretable timing, pausing, correction, rhythm, speed, and variability indicators. Retrained ablation additionally shows that strong standalone predictive value does not imply unique incremental contribution within fusion.

RQ4 exposes the central systems trade-off: complete fusion improved macro-F1 by **0.0097** over the near-equivalent 22-feature keystroke model, but required substantially greater dimensionality, computational cost, and input availability.

Overall, the project objectives were achieved at **proof-of-concept level**, with the strongest evidence concerning engineering integration and within-dataset discrimination.

5.10 Limitations and Improvements

The principal limitations concern four forms of validity. **Internal validity** is constrained by class-guided text, selected audiovisual sources, and repeated model comparison on limited data, which may produce optimistic estimates (Cawley and Talbot, 2010).

Construct validity is limited because the four classes are operational labels rather than validated psychological measurements. High separability therefore demonstrates prediction of experimental labels, not direct measurement of latent cognitive state.

External validity is restricted by participant-specific keystrokes, heterogeneous audiovisual provenance, and the absence of unseen-user or unseen-source evaluation.

Statistical validity is constrained by the ratio of 2,373 predictors to 309 records, the absence of an independent external test set, and ceiling-level performance that limits discrimination among high-performing configurations. Deterministic record linkage additionally establishes schema-consistent association rather than natural temporal synchrony.

Future work should prioritise naturally co-recorded multi-participant data, participant-aware validation, an untouched external test set, calibration, dimensionality reduction, and alternative fusion strategies. These extensions would move the evidence from **within-dataset engineering feasibility toward participant-independent behavioural validation** rather than merely increasing model complexity.

CHAPTER 6: CONCLUSION (885/1000 words)

6.1 Research Summary

This project investigated whether heterogeneous behavioural and interaction signals could be coordinated within a modular framework for near-real-time behavioural-state classification. SenseFuzeAI integrated keystroke dynamics, textual content, acoustic representations, and visual-semantic features to classify four operational states: **Focused**, **Distracted**, **Fatigued**, and **Overloaded**.

The implemented system comprised browser-based acquisition, modality-specific preprocessing and feature extraction, keystroke behavioural abstraction, deterministic record linkage, feature-schema validation, multimodal fusion, supervised classification, confidence presentation, and structured prediction logging. MPNet, WavLM, Librosa, and CLIP provided textual, acoustic, and visual representations, while engineered interaction features represented keystroke behaviour.

Evaluation used stratified five-fold cross-validation, supplementary hold-out analysis, confusion matrices, macro-averaged metrics, leave-one-modality-out ablation, synthetic missing-modality stress testing, repeated CPU latency benchmarking, and automated software verification. Keystroke Random Forest was the strongest unimodal model with mean macro-F1 of **0.9903**, while complete multimodal Random Forest achieved **1.0000**, an absolute improvement of **0.0097**. This modest gain does not establish statistically generalisable superiority of fusion.

The findings remain bounded by dataset construction. Multimodal records were class-aligned composites rather than naturally synchronous behavioural episodes; keystroke data originated from one participant; text was class-guided; and audio and visual samples had heterogeneous external provenance. The results therefore demonstrate within-dataset technical feasibility rather than population-level behavioural recognition.

6.2 Research Questions and Achievement of the Aim

RQ1 asked what software architecture could coordinate modality-specific models and record-aligned features to support near-real-time behavioural inference. The modular FastAPI architecture coordinated acquisition, modality-specific feature extraction, schema reconstruction, classification, and presentation, while the persisted **2,373-feature schema** maintained consistency between training and inference. Repeated CPU benchmarking measured mean warm end-to-end latency of **2,501.15 ms** and p95 latency of **3,830.72 ms**. These results support interactive bounded-sample inference on the evaluated hardware, although p95 exceeded the 2.5-second update cadence and therefore does not establish a strict real-time guarantee.

RQ2 asked how multimodal fusion compared with individual modalities in predictive performance and robustness. Complete fusion achieved macro-F1 of **1.0000**, compared with **0.9903** for the strongest unimodal model, an absolute improvement of **0.0097**. Leave-one-modality-out ablation showed the largest retrained-model decrease after removing text ($\Delta F1 = 0.0065$), while inference-time zero masking showed the greatest sensitivity to vision, reducing macro-F1 to **0.9190** ($\Delta F1 = 0.0810$). Keystroke masking reduced macro-F1 to **0.9737** ($\Delta F1 = 0.0263$), whereas text and audio masking caused no measurable degradation. Robustness was therefore modality- and evaluation-condition-dependent.

RQ3 asked how keystroke features could be transformed into interpretable indicators and what contribution they made to classification. Twenty-two engineered features representing timing, pausing, correction, rhythm, speed, and variability produced the strongest unimodal performance. Leave-keystroke-out ablation retained macro-F1 of **1.0000** ($\Delta F1 = 0.0000$), indicating no unique incremental contribution after retraining, whereas inference-time masking reduced macro-F1 to **0.9737**. Keystrokes therefore provided strong standalone evidence, while substantial redundancy existed within the retrained fused representation.

RQ4 examined trade-offs between predictive performance, computational responsiveness, robustness, and interpretability. Complete fusion maximised within-dataset performance but required **2,373 predictors**, compared with only **22 features** for the near-equivalent keystroke model. Its **0.0097** macro-F1 advantage therefore came with substantially greater dimensionality and mean warm processing latency of approximately **2.50 s**. Ablation demonstrated compensatory redundancy after retraining, while zero-masking stress testing revealed particular sensitivity to vision loss ($\Delta F1 = 0.0810$). The prototype therefore exhibits a measurable trade-off between predictive gain, processing cost, robustness, input requirements, and interpretability.

Overall, the project achieved its architectural and classification objectives at proof-of-concept level. Ablation, missing-information stress testing, and latency benchmarking strengthened the evidence for modality contribution, predictive sensitivity, and computational responsiveness. Participant-independent generalisation, external validation, probability calibration, and evaluation on naturally co-recorded multimodal behavioural episodes remain unresolved.

6.3 Principal Contributions

SenseFuzeAI contributes three principal elements. First, it provides a modular record-aligned orchestration architecture for coordinating heterogeneous behavioural modalities and pretrained representations. Second, it introduces an interpretable keystroke behavioural-abstraction layer that transforms low-level interaction measurements into inspectable indicators without treating them as deterministic psychological measurements. Third, it provides an empirical evaluation of unimodal and multimodal classification, modality contribution, missing-information sensitivity, and computational responsiveness within a common framework.

The web application additionally demonstrates practical integration of acquisition, inference, class probabilities, confidence-gap reporting, modality-status information, and prediction logging.

6.4 Limitations and Future Work

The principal limitations concern construct validity, dataset provenance, generalisability, and dimensionality. The four behavioural states are operational labels rather than direct psychological measurements, while class-guided text, externally sourced audio-visual material, and participant-specific keystrokes may introduce dataset-specific correlations. The **2,373-predictor model trained on 309 class-aligned composite records** also presents overfitting risk despite leakage-safe preprocessing and cross-validation. Results should therefore be interpreted as within-dataset proof-of-concept evidence rather than participant-independent behavioural recognition.

Leave-one-modality-out ablation, classifier-level zero-masking stress testing, and latency benchmarking strengthen evidence for modality contribution, predictive sensitivity, and responsiveness, but do not establish external validity. Future work should prioritise naturally co-recorded multimodal data from larger participant populations, group-aware or leave-one-participant-out validation, independent external validation, probability calibration, dimensionality reduction, and alternative fusion strategies.

Future deployment should also preserve privacy, informed consent, secure data retention, data minimisation, transparency, and explicit user control over webcam, microphone, and interaction-data acquisition.

6.5 Final Conclusion Remarks

SenseFuzeAI demonstrates that heterogeneous behavioural and semantic representations can be coordinated within an operational multimodal inference pipeline. Final automated verification executed 158 pytest cases successfully. The canonical project-wide runner additionally completed 41 verification checks with zero failures or warnings; one optional heavyweight model-initialisation smoke check was intentionally skipped.

The project therefore establishes a critically evaluated foundation for future participant-independent, calibrated, privacy-conscious, and empirically validated multimodal behavioural research.

CHAPTER 7: APPENDICES

Appendix A: Project Artefacts and Reproducibility

The following artefacts support inspection, demonstration, and reproduction of the SenseFuzeAI prototype.

A.1 Source-Code Repository

GitHub repository:

<https://github.com/ngclara07/CM3070-Final-Project.git>

The repository contains the source code for multimodal data acquisition, modality-specific feature extraction, classifier training and evaluation, feature-level fusion, the FastAPI web application, and automated software testing. Supporting files include the project dependencies, persisted feature schemas, evaluation scripts, and setup instructions required to reproduce the demonstration environment.

The repository will remain publicly accessible for assessment in accordance with the project-submission requirements.

A.2 Demonstration dataset:

The repository includes a 20-record demonstration dataset generated for lightweight software inspection and demonstration. This dataset is separate from the 309-record experimental dataset used for the evaluation reported in Chapter 5.

The experimental dataset comprised 77 Focused, 77 Distracted, 77 Fatigued, and 78 Overloaded composite records. The smaller demonstration dataset reproduces the expected data structure and processing workflow but is not intended to reproduce the complete quantitative results reported in Chapter 5.

A.3 Pretrained models:

Large pretrained model weights are not stored directly in the repository because of their size. Setup instructions in README.md specify how the required pretrained models are obtained from their original distribution sources.

The principal pretrained models are:

- sentence-transformers/all-mpnet-base-v2 for textual representation;
- microsoft/wavlm-base-plus for acoustic representation; and
- openai/clip-vit-large-patch14 for visual representation.

A.4 Installation and Execution

Installation, dependency configuration, model setup, dataset preparation, training, evaluation, and application-execution instructions are provided in the repository README.md.

The principal environment and reproducibility settings used for the reported experiments are documented in Section 5.2.

A.5 Demonstration Video

Project demonstration video:

The project demonstration video is submitted separately in MP4 format in accordance with the assessment requirements. For alternative access, the video is also available via Google Drive:

Video URL:

<https://drive.google.com/file/d/1vFlsESxBhVqELrkSegKhvHdBbUwe3Xc0/view?usp=sharing>

The demonstration presents the principal end-to-end workflow of SenseFuzeAI, including multimodal input acquisition, behavioural-state inference, probability and confidence presentation, modality-status information, and prediction logging.

Appendix B: Dataset Selection and Supplementary Keystroke Comparison Evidence

B.1 Purpose and Scope

An exploratory investigation of the IEEE EmoSurv keystroke dataset preceded development of the final SenseFuzeAI evaluation dataset. EmoSurv provides multi-participant keystroke observations labelled using emotional states, whereas SenseFuzeAI addresses the operational behavioural classes Focused, Distracted, Fatigued, and Overloaded. The exploratory work was therefore used to assess the feasibility of keystroke-based behavioural classification and to inform subsequent dataset design rather than as the principal evidence for the final multimodal system.

The final SenseFuzeAI evaluation instead used 309 class-aligned composite records comprising 77 Focused, 77 Distracted, 77 Fatigued, and 78 Overloaded records. The shared session_id supports deterministic linkage between modality-specific feature tables but does not imply simultaneous acquisition, common participant identity, or a naturally co-recorded behavioural episode.

A later supplementary comparison pipeline harmonised EmoSurv and SenseFuzeAI keystroke observations to a common feature representation. This comparison provides evidence concerning dataset compatibility, source-domain differences, and group-safe evaluation. It is supplementary to, and should not be interpreted as replacing, the principal multimodal evaluation reported in Chapter 5.

B.2 Exploratory EmoSurv Evaluation

The IEEE EmoSurv dataset contains fixed-text and free-text keystroke records originally labelled angry, calm, happy, neutral, and sad. These labels do not correspond directly to the four operational SenseFuzeAI states. During the exploratory investigation, behavioural proxy labels were therefore derived from the original emotion labels. Because this transformation required heuristic mapping, the resulting targets were treated as weak behavioural proxies rather than direct behavioural-state ground truth.

The preprocessing pipeline generated 1,837 non-overlapping keystroke windows from 83 participants, represented using 23 keystroke-dynamics features. A participant-independent grouped partition produced 1,387 training windows from 66 participants and 450 held-out windows from 17 participants, with no participant appearing in both partitions.

Table B.1. Exploratory EmoSurv classifier comparison

Model	Accuracy	Macro Precision	Macro Recall	Macro F1	Weighted F1
Baseline Random Forest	74.44%	78.50%	51.98%	52.14%	69.01%
Class-weighted Random Forest	72.89%	69.38%	52.64%	54.62%	69.06%

The class-weighted Random Forest was retained because it achieved the higher macro-F1 score. Its held-out accuracy was 72.89% and macro-F1 was 54.62%. Class-level performance was uneven: recall was 23.26% for Fatigued and 16.67% for Overloaded, demonstrating that overall accuracy overstated consistency across the generated proxy classes.

The held-out partition was used to compare the two Random Forest configurations and should therefore be interpreted as a model-comparison partition rather than a completely untouched post-selection test set. The experiment demonstrated that the extracted keystroke features contained information associated with the generated behavioural proxies, but also exposed limitations arising from proxy labelling, class imbalance, and construct mismatch.

B.3 Rationale for the Final SenseFuzeAI Dataset

EmoSurv offered two important advantages: an externally collected keystroke source and observations from multiple participants. However, it was unsuitable as the principal dataset for the final SenseFuzeAI multimodal experiment for two reasons. First, its original emotional labels required indirect mapping to the operational behavioural classes. Second, it contained keystroke observations only and therefore provided no corresponding text, audio, or visual records for the intended feature-level multimodal fusion task.

The final SenseFuzeAI dataset was constructed around the four project-defined behavioural labels and provided modality-specific records that could be combined deterministically into a common classification schema. The resulting dataset contained 309 complete class-aligned composite records. Text was manually authored under class-guided conditions, keystrokes originated from one participant, and audio and visual samples were obtained from external online sources. Consequently, the final dataset should not be described as naturally synchronous or participant-matched multimodal data.

Dataset selection was therefore based primarily on alignment with the research questions, label compatibility, and support for multimodal feature construction, rather than on which dataset produced the highest predictive score. The final dataset provides an appropriate basis for a controlled proof-of-concept evaluation, while its participant-specific and heterogeneous provenance remains an explicit limitation.

Table B.2. Methodological comparison of the two dataset approaches

Criterion	EmoSurv exploratory dataset	Final SenseFuzeAI evaluation dataset
Original target labels	Emotion labels	Focused, Distracted, Fatigued, Overloaded
Keystroke participants	83	One participant
Available modalities	Keystroke only	Keystroke, text, audio, vision
Label relationship to project	Requires behavioural proxy mapping	Direct project operational classes
Multimodal use	No corresponding non-keystroke modalities	Supports deterministic class-aligned fusion
Principal strength	Multi-participant external keystroke evidence	Alignment with the intended multimodal task

Principal limitation	Construct mismatch and no multimodal correspondence	Limited participant diversity and heterogeneous source provenance
Role in final project	Exploratory/supporting evidence	Principal Chapter 5 evaluation dataset

B.4 Harmonised EmoSurv-SenseFuzeAI Keystroke Comparison

A later dataset-comparison pipeline was implemented to provide a more controlled comparison between the two keystroke sources. Rather than comparing incompatible feature tables directly, both datasets were transformed to an exact 23-feature harmonised keystroke schema covering dwell-time statistics, key-transition timing, typing speed, pause ratios, correction and space ratios, rhythm variability, and overlap behaviour. Metadata such as source identity, participant identifiers, session identifiers, sample identifiers, and label provenance were preserved separately and excluded from the classifier feature schema.

This harmonised 23-feature representation is specific to the supplementary dataset-comparison pipeline. It should not be confused with the 22 engineered keystroke predictors used by the principal SenseFuzeAI multimodal classifier reported in Chapter 5.

The comparison framework defines a three-class primary analysis using Focused, Fatigued, and Overloaded, together with a four-class exploratory analysis additionally including Distracted. It contains six canonical experiment configurations:

- A_emosurv_baseline
- B_sensefuzeai_only
- C_augmented_to_emosurv_test
- D_emosurv_to_sensefuzeai
- E_augmented_to_sensefuzeai_test
- F_sensefuzeai_to_emosurv

The implementation uses frozen group-aware split manifests rather than unrestricted row-level random splitting. EmoSurv participant identifiers are separated between training and test groups, while SenseFuzeAI session/group identifiers are similarly checked for overlap. Baseline and augmented comparisons are also required to use identical held-out samples where direct paired comparison is intended.

The comparison outputs report accuracy, balanced accuracy, and macro-F1, and the trainer additionally contains bootstrap uncertainty analysis. Automated verification checks the harmonised schema, dataset provenance, numeric validity, source separation, frozen splits, leakage prevention, matched held-out samples, and presence of all six comparison configurations.

The associated “test_05_keystroke_dataset_comparison.py” module does not retrain the comparison models. It verifies the source contracts and already-generated comparison artefacts, preserving a separation between software verification and experimental model evaluation.

B.5 Interpretation

The EmoSurv and SenseFuzeAI datasets should not be treated as directly interchangeable benchmarks. They differ in original labels, participant composition, observation construction, collection procedure, and intended role within the project. Consequently, differences in predictive performance cannot be attributed solely to classifier quality.

EmoSurv contributes useful external, multi-participant keystroke evidence but relies on behavioural proxy mapping when adapted to the SenseFuzeAI task. The final SenseFuzeAI dataset provides direct alignment with the four operational project labels and supports multimodal fusion, but its keystroke component is participant-specific and the fused records are constructed from heterogeneous modality sources rather than naturally co-recorded behavioural episodes.

The supplementary harmonised comparison therefore serves primarily as supporting evidence about keystroke representation compatibility, source-domain behaviour, and evaluation methodology. The principal claims of the project remain those established by the 309-record multimodal evaluation in Chapter 5. Neither the exploratory EmoSurv experiment nor the harmonised comparison removes the stated limitations concerning construct validity, participant-independent generalisation, or naturally synchronous multimodal acquisition.

Appendix C: Selected Implementation and Verification Evidence

C.1 Purpose and Scope

This appendix provides selected source-code and execution-verification evidence supporting the software-engineering claims discussed in Chapters 4 and 5. The excerpts are intentionally limited to architectural mechanisms central to SenseFuzeAI: shared temporal aggregation, multimodal feature-schema consistency, and reproducible automated verification. Complete source code, test modules, model artefacts, and execution instructions are available through the public repository identified in Appendix A.

The excerpts are illustrative rather than exhaustive. They demonstrate how selected architectural properties reported in the main text are enforced programmatically without reproducing substantial portions of source code already available in the repository.

C.2 Shared Temporal-Fusion Contract

SenseFuzeAI separates stateless raw multimodal inference from temporal aggregation. A shared TemporalFusionEngine provides the canonical temporal behaviour used by application-level consumers, while FinalMultimodalInference remains responsible for producing the raw four-class probability representation. Automated acceptance testing verifies that the canonical temporal probability window is configured to contain five observations.

Listing C.1. Acceptance-test excerpt verifying the canonical temporal-fusion configuration (tests/test_04_acceptance.py)

```
def test_temporal_fusion_is_single_source_of_truth():  
  
    from temporal_fusion import (  
        LABELS,  
        TEMPORAL_PROBABILITY_WINDOW,  
    )  
  
    assert tuple(LABELS) == EXPECTED_LABELS  
  
    assert (
```

```
    TEMPORAL_PROBABILITY_WINDOW
    == 5
)
```

Source: Author-developed test suite. Selected source-code excerpt; formatting condensed for presentation without altering the tested logic.

The temporal engine is also maintained independently for each web session. The integration test below constructs two session states, appends one probability observation to the first session, and verifies that the second session remains unaffected. This prevents temporal prediction history from leaking between concurrent logical sessions.

Listing C.2. Integration-test excerpt verifying isolation of temporal state across web sessions (tests/test_02_integration.py)

```
def test_web_session_state_has_independent_temporal_engines(
    app_module: ModuleType,
):
    first = app_module.SessionState()
    second = app_module.SessionState()

    assert (
        first.temporal_fusion
        is not second.temporal_fusion
    )

    first.temporal_fusion.append({
        "focused": 1.0,
        "distracted": 0.0,
        "fatigued": 0.0,
        "overloaded": 0.0,
    })

    assert first.temporal_fusion.sample_count == 1
    assert second.temporal_fusion.sample_count == 0
```

Source: Author-developed test suite. Selected source code excerpt; formatting condensed for presentation without altering the tested logic.

The canonical project-wide runner additionally verifies probability normalisation, invalid-probability handling, confidence thresholds, five-sample arithmetic averaging, rolling-window rollover, reset and stale-generation behaviour, consumer parity, and probability validation. These checks provide broader verification of temporal behaviour than the two representative excerpts shown above.

C.3 Fusion Schema and Inference Contract

The deployed fusion classifier expects a fixed persisted feature schema. The final representation contains 2,373 predictors, comprising 22 keystroke features, 768 textual features, 809 audio features, 768 visual features, and six additional derived predictors. The production build_fusion_dataframe operation reconstructs the persisted 2,373-dimensional fusion row before classification.

Integration testing verifies that the number of predictors expected by the fitted classifier is consistent with the persisted feature schema, reducing the risk of silent incompatibility between training and inference.

Listing C.3. Integration-test excerpt verifying consistency between the trained fusion model and persisted feature schema (tests/test_02_integration.py)

```
def test_fusion_model_feature_count_matches_schema_when_available():

    model = joblib.load(
        FUSION_MODEL_PATH
    )

    columns = json.loads(
        FUSION_SCHEMA_PATH.read_text(
            encoding="utf-8"
        )
    )

    feature_count = getattr(
        model,
        "n_features_in_",
        None,
    )
```

```
if feature_count is not None:
    assert int(feature_count) == len(columns)
```

Source: Author-developed test suite. Selected source-code excerpt; formatting condensed for presentation without altering the tested logic.

The inference architecture is separately checked for the required modality-processing and fusion methods. In particular, the integration contract verifies that `build_fusion_dataframe`, the modality-specific feature-extraction methods, and the final `predict` interface remain available through `FinalMultimodalInference`.

Listing C.4. Integration-test excerpt verifying the required public API of `FinalMultimodalInference` (`tests/test_02_integration.py`)

```
required_methods = {
    "build_fusion_dataframe",
    "extract_audio_features",
    "extract_clip_embedding",
    "extract_image_features",
    "extract_keystroke_features",
    "extract_text_features",
    "extract_webcam_calibration_features",
    "predict",
}

for method in required_methods:
    assert hasattr(
        inference_class,
        method,
    )
```

Source: Author-developed test suite. Selected source-code excerpt; surrounding test setup omitted and formatting condensed without altering the tested logic.

These checks complement the schema-validation behaviour described in Chapter 4. Together, they verify structural compatibility between the trained model, persisted feature schema, modality-specific feature-extraction interfaces, and final fusion-inference component following implementation changes.

C.4 Final Automated Verification Evidence

The final project-wide verification was executed using the canonical test runner.

Command C.1. Canonical project-wide verification command

```
python run_all_tests.py --require-pytest --strict --verbose
```

The runner completed with the following summary.

Output C.1. Summary produced by the canonical project-wide verification runner

```
PASS: 41  
FAIL: 0  
WARN: 0  
SKIP: 1
```

```
FINAL RESULT: PASS
```

Source: Author-generated terminal output from the final project-wide verification run.

The runner incorporates Python syntax checking, canonical temporal mathematics, architectural contracts, continuous-browser-audio behaviour, keystroke dataset-comparison contracts, JavaScript validation, and the complete pytest suite. It also distinguishes mandatory failures, strict-mode warnings, and intentionally optional checks before determining the final process result.

The embedded pytest stage produced:

Output C.2. Embedded pytest execution result

```
158 passed in 13.45s
```

Source: Author-generated terminal output from the canonical project-wide verification run.

An independent pytest execution was subsequently performed using:

Command C.2. Independent pytest verification command

```
python -m pytest tests/ -q
```

which reproduced the result:

Output C.3. Independent pytest reproduction

```
158 passed in 7.17s
```

Source: Author-generated terminal output from the independent pytest verification run.

The 41 project-wide checks and 158 pytest cases should not be summed as 199 independent tests, because the canonical runner incorporates the complete pytest execution as one of its project-wide verification stages. The single reported skip corresponded to the optional heavyweight FinalMultimodalInference initialisation smoke check, which is disabled unless explicitly enabled and therefore did not represent a failed mandatory verification. This interpretation is consistent with the software-testing results reported in Section 5.8.

C.5 Interpretation

The selected excerpts demonstrate three implementation properties central to the final system: temporal aggregation is governed through a shared and session-isolated mechanism; the deployed multimodal classifier remains coupled to an explicit persisted feature schema; and the principal architectural assumptions are exercised through a repeatable automated verification process.

These excerpts provide supporting evidence of implementation consistency and software correctness. They do not independently establish behavioural construct validity, participant-independent predictive generalisation, or reliability across arbitrary deployment hardware. Those limitations remain addressed through the experimental evaluation and critical discussion in Chapter 5.

Appendix D: Participant Information and Consent Form

D.1 Purpose

This appendix reproduces the blank participant information and consent form used for the scenario-based screenshot usability evaluation of SenseFuzeAI. Signed participant copies are excluded from the dissertation and retained privately. The form explains the evaluation purpose, voluntary participation, anonymised use of responses, and that no webcam, microphone, keystroke, or behavioural recording data are collected during the study.

D.2 Blank Participant Information and Consent Form

SenseFuzeAI Participant Information & Consent Form

Scenario-Based Screenshot Usability Evaluation

Project: SenseFuzeAI – Multimodal Behavioural AI System

Study Type: Undergraduate Computer Science Final Project

Evaluation Type: Formative Scenario-Based Screenshot Usability Evaluation

About This Evaluation

You are invited to take part in a short usability evaluation of the SenseFuzeAI web application interface.

SenseFuzeAI is an academic artificial-intelligence system that combines information from text, keystroke, audio, and visual inputs to produce a behavioural-state prediction.

The purpose of this evaluation is to obtain independent feedback on the:

- clarity of the interface
- visual layout and organisation
- understandability of controls and status information
- presentation of behavioural-state predictions
- presentation of confidence information
- overall perceived usability of the application

Your feedback will be used to evaluate the SenseFuzeAI interface and identify possible improvements as part of an undergraduate Computer Science final project.

What Will I Be Asked to Do?

You will be shown screenshots representing different stages of the SenseFuzeAI workflow.

These may include:

- system readiness
- multimodal controls
- session controls
- behavioural-state prediction results
- confidence information
- technical diagnosis information

You will answer a short series of multiple-choice, rating-scale, and open-ended questions based on what you can see and understand from the screenshots.

Expected completion time: approximately 10-15 minutes.

Important

This is a screenshot-based usability evaluation.

You will not be required to directly operate the live SenseFuzeAI application as part of this evaluation.

What Information Will Be Collected?

The questionnaire may collect:

- your general confidence using web applications
- multiple-choice responses
- usability ratings
- written comments about the interface

The questionnaire does not require your name, student ID, participant ID, or email address.

The evaluation will also not collect:

- webcam images
- microphone recordings
- keystroke samples
- behavioural recordings
- or other multimodal input data from you

Privacy and Anonymity

Your questionnaire responses will be collected anonymously.

Your responses may be summarised in the final academic project or dissertation. For example, the report may state:

- “One respondent found the distinction between the two reset controls unclear”

Individual participants will not be personally identified in the reported usability findings.

Because responses are collected anonymously, once a submitted response can no longer be linked to an individual participant, it may not be possible to remove that individual response from the study.

How Will My Feedback Be Used?

Your responses may be used to:

- evaluate the usability of the SenseFuzeAI interface
- identify areas of confusion
- identify possible improvements to labels, controls, layout, or presentation
- support discussion of user-centred system development
- provide evidence of formative usability evaluation in the final project/dissertation

Any interpretation of the results will acknowledge that this is a small formative evaluation and is not intended to represent all potential SenseFuzeAI users statistically.

Voluntary Participation

Taking part is entirely voluntary.

You may:

- choose not to participate
- decline to answer an optional question
- or stop the evaluation at any time before submission your responses

There is no penalty for choosing not to participate.

You are not being tested on your technical knowledge or ability.

The purpose of the evaluation is to assess the interface, not the participant.

Possible Risks and Benefits

This is a low-risk usability evaluation.

No significant risks are expected beyond the small amount of time required to review the screenshots and complete the questionnaire.

Some technical terminology may be unfamiliar. You are not expected to have specialist knowledge of artificial intelligence or machine learning.

There may be no direct personal benefit from taking part. However, your feedback may help identify ways to improve the clarity and usability of the SenseFuzeAI interface.

Consent

Please read the following statements before returning to the questionnaire.

By choosing “**Yes, I agree to participate**” in the Google Form, you confirm that:

- ☐ you have read and understood the information provided in this document;
- ☐ you understand the purpose of the usability evaluation;
- ☐ you understand that you will review screenshots of the SenseFuzeAI interface and answer usability questions;

- ☐ you understand that you will not directly operate the live SenseFuzeAI application during this evaluation;
- ☐ you understand that participation is voluntary;
- ☐ you understand that your questionnaire responses will be collected anonymously;
- ☐ you understand that anonymised responses may be used in the final academic project/dissertation;
- ☐ you understand that no webcam, microphone, keystroke, or behavioural recording data will be collected from you; and
- ☐ you voluntarily agree to participate

Anonymous Written Comments

The questionnaire may include open-ended questions asking for your opinions about the interface. Short anonymous quotations from written feedback may be included in the final academic project/dissertation only where you give permission in the questionnaire.

Appendix E: SenseFuzeAI Scenario-Based Screenshot Usability Evaluation

E.1 Purpose

This appendix reproduces the blank questionnaire used for the formative scenario-based usability evaluation of the SenseFuzeAI interface. Participants reviewed screenshots representing system readiness, multimodal input controls, behavioural-state prediction, and technical diagnostics before completing structured and open-ended usability questions. Only the blank questionnaire is reproduced; participant responses are reported anonymously where relevant.

E.2 Blank Scenario-Based Screenshot Usability Questionnaire

SenseFuzeAI Scenario-Based Screenshot Usability Evaluation

Formative Interface Comprehension and Perceived Usability Study

Introduction

Have you previously seen or used SenseFuzeAI before this evaluation?

- ☐ Yes
☐ No

How would you describe your general confidence when using web applications?

- ☐ Low
☐ Moderate
☐ High

Section 1 – System Readiness

Scenario

Imagine that you have just opened SenseFuzeAI.

The system has loaded its AI models but requires some user inputs before it can produce a behavioural-state prediction.

Screenshot 1 – Live System Readiness

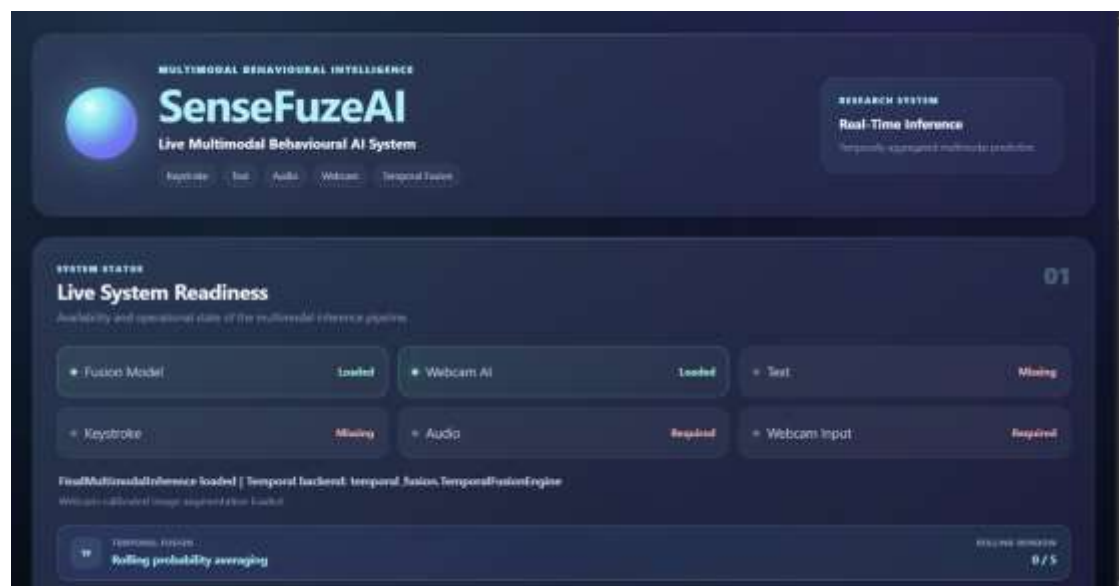


Figure 1: SenseFuzeAI Live System Readiness interface

Q1. From this screen, is it clear whether SenseFuzeAI is currently ready to make a prediction?

- ☐ Yes
- ☐ Partly
- ☐ No

Q2. Which inputs appear to require attention before a prediction can be produced? Tick all that apply.

- ☐ Text
- ☐ Keystroke
- ☐ Audio
- ☐ Webcam / visual input
- ☐ Fusion model
- ☐ Webcam AI
- ☐ Not sure

Q3. Is the distinction between Loaded, Missing, and Required understandable?

- ☐ Yes
- ☐ Partly
- ☐ No

Q4. How clear is the Live System Readiness section?

- ☐ 1 – Very unclear
- ☐ 2 – Unclear
- ☐ 3 – Neutral
- ☐ 4 – Clear
- ☐ 5 – Very clear

Q5. If you were using the application at this point, would you know what to do next?

- ☐ Yes
- ☐ Not sure
- ☐ No

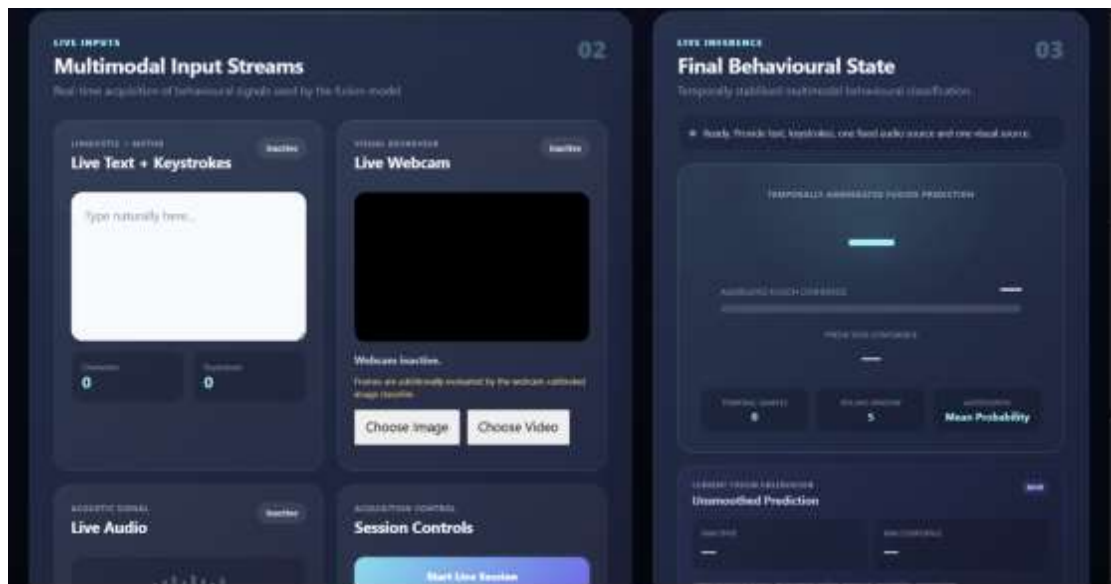
Optional comment – Was anything unclear?

Section 2 – Inputs and Session Controls

Scenario

You now want to provide the information required by SenseFuzeAI and begin a live prediction session.

Screenshot 2 – Multimodal Inputs and Controls



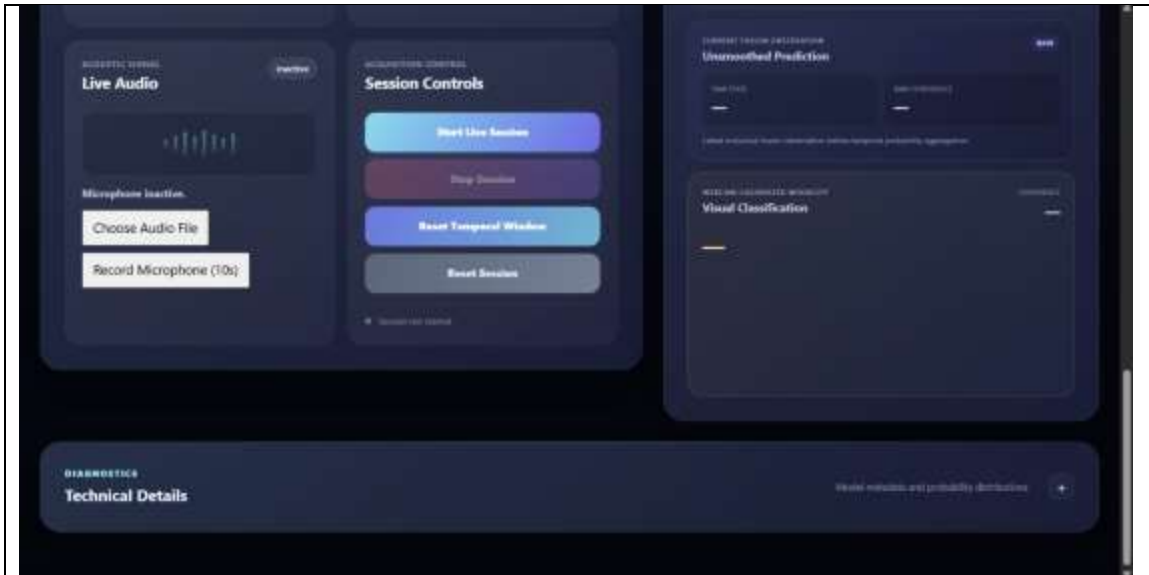


Figure 2: SenseFuzeAI multimodal input and live-session controls

Q6. Is it clear where you would provide text and keystroke input?

- ☐ Yes
- ☐ Partly
- ☐ No

Q7. Is it clear how you would provide an audio source?

- ☐ Yes
- ☐ Partly
- ☐ No

Q8. Is it clear how you would provide a webcam, image or video source?

- ☐ Yes
- ☐ Partly
- ☐ No

Q9. Which control would you expect to use to begin live inference?

- ☐ Start Live Session
- ☐ Stop Session
- ☐ Reset Temporal Window
- ☐ Reset Session
- ☐ Not sure

Q10. Which control would you use to clear only the recent temporal prediction history?

- ☐ Start Live Session
- ☐ Stop Session
- ☐ Reset Temporal Window
- ☐ Reset Session
- ☐ Not sure

Q11. Which control would you use to restart the whole interaction from the beginning?

- ☐ Start Live Session
- ☐ Stop Session
- ☐ Reset Temporal Window
- ☐ Reset Session
- ☐ Not sure

Q12. How easy does this section appear to navigate?

- ☐ 1 – Very difficult
- ☐ 2 – Difficult
- ☐ 3 – Neutral
- ☐ 4 – Easy
- ☐ 5 – Very easy

Optional comment – Is any button or control unclear?

Section 3 – Behavioural-State Prediction

Scenario

SenseFuzeAI has now received the required inputs and produced a behavioural-state prediction.

Screenshot 3 – Final Behavioural State

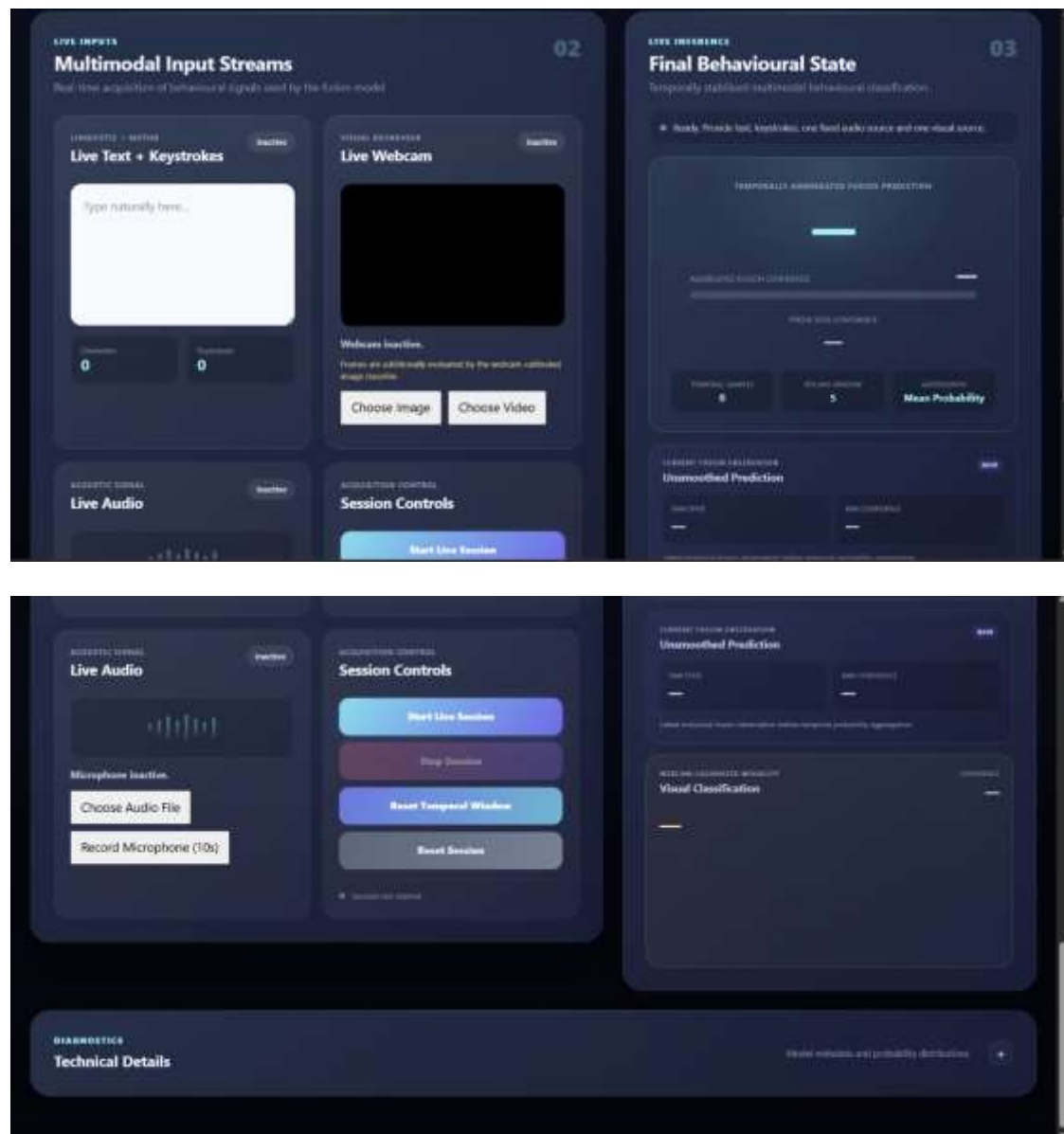


Figure 3: SenseFuzeAI final behavioural-state prediction and confidence information

Q13. What do you identify as the main result displayed by SenseFuzeAI?

Q14. Is the main behavioural-state prediction easy to locate?

- ☐ Yes
☐ Partly
☐ No

Q15. Is the confidence information easy to understand?

- ☐ Yes
☐ Partly
☐ No

Q16. Is it clear that the main result represents a **temporally aggregated prediction**?

- ☐ Yes
☐ Partly
☐ No

Q17. Is the difference between the final temporally aggregated prediction and the **raw / unsmoothed prediction** understandable?

- ☐ Yes
☐ Partly
☐ No

Q18. How clear is the Final Behavioural State section overall?

- ☐ 1 – Very unclear
☐ 2 – Unclear
☐ 3 – Neutral
☐ 4 – Clear
☐ 5 – Very clear

Optional comment – Would you change anything about the prediction or confidence information?

Section 4 – Technical Details

Scenario

SenseFuzeAI provides an expandable Technical Details area for users who want to inspect additional system information.

Screenshot 4 – Technical Diagnostics

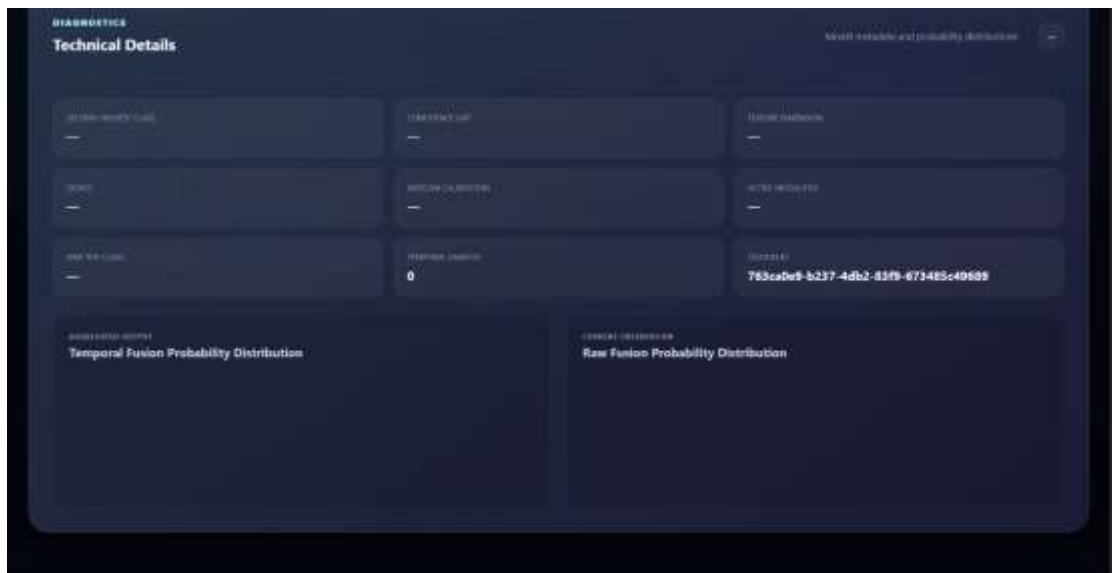


Figure 4: SenseFuzeAI expanded technical diagnostics and probability distributions

Q19. Is it clear that this section contains additional technical information rather than the main user-facing result?

- ☐ Yes
☐ Partly
☐ No

Q20. Is it clear that the Technical Details section can be expanded or collapsed?

- ☐ Yes
- ☐ Partly
- ☐ No

Q21. Do the probability-distribution displays appear understandable?

- ☐ Yes
- ☐ Partly
- ☐ No

Q22. How does the amount of technical information feel?

- ☐ Too little
- ☐ About right
- ☐ Too much
- ☐ Not sure

Q23. If you were using SenseFuzeAI, would you want to view the Technical Details section?

- ☐ Yes
- ☐ Sometimes
- ☐ No

Q24. How well is the technical information separated from the main prediction?

- ☐ 1 – Very poorly
- ☐ 2 – Poorly
- ☐ 3 – Adequately
- ☐ 4 – Well
- ☐ 5 – Very well

Optional comment – Was any technical term or display difficult to understand?

Section 5 – Overall Interface Evaluation

Please consider all four screenshots.

Q25. Overall, the SenseFuzeAI interface appears easy to understand and navigate.

- ☐ 1 – Strongly Disagree
- ☐ 2 – Disagree
- ☐ 3 – Neutral
- ☐ 4 – Agree
- ☐ 5 – Strongly Agree

Q26. Overall, how would you rate the visual/interface design of SenseFuzeAI?

- ☐ 1 – Very poor
- ☐ 2 – Poor
- ☐ 3 – Acceptable
- ☐ 4 – Good
- ☐ 5 – Excellent

Q27. Based on the screenshots, how confident are you that you would know how to begin using SenseFuzeAI without detailed instructions?

- ☐ 1 – Not confident at all
 - ☐ 2 – Slightly confident
 - ☐ 3 – Moderately confident
 - ☐ 4 – Confident
 - ☐ 5 – Very confident
-

Section 6 – Open-Ended Feedback

Q28. What part of the SenseFuzeAI interface appears most confusing or difficult to understand?

Q29. What part of the interface appears clearest or easiest to understand?

Q30. If you could improve **one thing** about the SenseFuzeAI interface, what would you change and why?

Q31. Based on the screenshots and scenarios shown, do you think you would enjoy using the SenseFuzeAI application? Why or why not?

Q32. Based on what you have seen, would you recommend that other people try the SenseFuzeAI application? Why or why not?

Thank You

Thank you for participating in the **SenseFuzeAI Scenario-Based Screenshot Usability Evaluation**.

Your feedback will be used to identify possible improvements to the clarity, presentation and interaction design of the SenseFuzeAI web application.

E.3 Online Questionnaire Access

An online version of the blank questionnaire was implemented using Google Forms:

<https://forms.gle/w52hcLTEJ3suuCDc8>

CHAPTER 8: REFERENCES

1. Acien, A., Morales, A., Vera-Rodriguez, R., Fierrez, J., Mondesire-Crump, I. and Arroyo-Gallego, T. (2022) 'Detection of mental fatigue in the general population: Feasibility study of keystroke dynamics as a real-world biomarker', *JMIR Biomedical Engineering*, 7(2), e41003. doi: 10.2196/41003.
2. Baltrušaitis, T., Zadeh, A., Lim, Y.C. and Morency, L.-P. (2018) 'OpenFace 2.0: Facial behavior analysis toolkit', in *Proceedings of the 13th IEEE International Conference on Automatic Face and Gesture Recognition*, pp. 59-66. doi: 10.1109/FG.2018.00019.
3. Baltrušaitis, T., Ahuja, C. and Morency, L.-P. (2019) 'Multimodal machine learning: A survey and taxonomy', *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 41(2), pp. 423-443. doi: 10.1109/TPAMI.2018.2798607.
4. Bennett, D., Roudaut, A. and Metatla, O. (2025) 'Multifractality in typing as a marker of fatigue', *International Journal of Human-Computer Studies*, 204, 103595. doi: 10.1016/j.ijhcs.2025.103595.
5. Cawley, G.C. and Talbot, N.L.C. (2010) 'On over-fitting in model selection and subsequent selection bias in performance evaluation', *Journal of Machine Learning Research*, 11, pp. 2079-2107.
6. Chen, S. et al. (2022) 'WavLM: Large-scale self-supervised pre-training for full stack speech processing', *IEEE Journal of Selected Topics in Signal Processing*, 16(6), pp. 1505-1518. doi: 10.1109/JSTSP.2022.3188113.
7. Dutta, S., Balaji, S. and Ganapathy, S. (2026) 'A mixture-of-experts model for multimodal emotion recognition in conversations', *Computer Speech & Language*, 100, 101965. doi: 10.1016/j.csl.2026.101965.
8. Epp, C., Lippold, M. and Mandryk, R.L. (2011) 'Identifying emotional states using keystroke dynamics', in *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems*, pp. 715-724. doi: 10.1145/1978942.1979046.
9. FastAPI (n.d.) *FastAPI documentation*. Available at: <https://fastapi.tiangolo.com/> (Accessed: 18 July 2026).
10. Gao, Z., Jiang, X., Xu, X., Shen, F., Li, Y. and Shen, H.T. (2024) 'Embracing unimodal aleatoric uncertainty for robust multimodal fusion', in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pp. 26876-26885.

11. He, J., Mi, J. and Toda, T. (2025) ‘GIA-MIC: Multimodal emotion recognition with gated interactive attention and modality-invariant learning constraints’, in *Proceedings of INTERSPEECH 2025*, pp. 2695-2699. doi: 10.21437/Interspeech.2025-2696.
12. Huang, J., Zhong, J., Lei, Q., Gao, J., Yang, Y., Wang, S., Li, P. and Wei, K. (2025) ‘Latent distribution decouple for uncertain-aware multimodal multi-label emotion recognition’, in *Findings of the Association for Computational Linguistics: ACL 2025*, pp. 24123-24138. doi: 10.18653/v1/2025.findings-acl.1238.
13. Maalej, A., Kallel, I. and Sánchez, M.J. (2020) *EmoSurv: A typing biometric (keystroke dynamics) dataset with emotion labels created using computer keyboards*. IEEE DataPort. doi: 10.21227/eae6-pk42. Available at: <https://ieee-dataport.org/open-access/emosurv-typing-biometric-keystroke-dynamics-dataset-emotion-labels-created-using> (Accessed: 19 April 2026).
14. McFee, B., Raffel, C., Liang, D., Ellis, D.P.W., McVicar, M., Battenberg, E. and Nieto, O. (2015) ‘librosa: Audio and music signal analysis in Python’, in *Proceedings of the 14th Python in Science Conference*, pp. 18-24. doi: 10.25080/Majora-7b98e3ed-003.
15. Paszke, A. et al. (2019) ‘PyTorch: An imperative style, high-performance deep learning library’, *Advances in Neural Information Processing Systems*, 32, pp. 8024-8035.
16. Pedregosa, F. et al. (2011) ‘Scikit-learn: Machine learning in Python’, *Journal of Machine Learning Research*, 12, pp. 2825-2830.
17. Radford, A., Kim, J.W., Hallacy, C., Ramesh, A., Goh, G., Agarwal, S., Sastry, G., Askell, A., Mishkin, P., Clark, J., Krueger, G. and Sutskever, I. (2021) ‘Learning transferable visual models from natural language supervision’, in *Proceedings of the 38th International Conference on Machine Learning*, PMLR 139, pp. 8748-8763.
18. Reimers, N. and Gurevych, I. (2019) ‘Sentence-BERT: Sentence embeddings using Siamese BERT-networks’, in *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing*, pp. 3982-3992. doi: 10.18653/v1/D19-1410.
19. Schütz, L., Cherif, Y., Sayffaerth, C., Weber, T. and Chiossi, F. (2026) ‘Typing behavior in human-LLM interaction: Keystroke dynamics reveal cognitive effort during prompting’, *Proceedings of the ACM on Human-Computer Interaction*. doi: 10.1145/3821685.

20. Sentence Transformers (n.d.) *all-mpnet-base-v2 model card*. Available at: <https://huggingface.co/sentence-transformers/all-mpnet-base-v2> (Accessed: 18 July 2026).
21. Shadman, R., Wahab, A.A., Manno, M., Lukaszewski, M., Hou, D. and Hussain, F. (2025) 'Keystroke dynamics: Concepts, techniques, and applications', *ACM Computing Surveys*, 57(11), Article 283, pp. 1-35. doi: 10.1145/3733103.
22. Sokolova, M. and Lapalme, G. (2009) 'A systematic analysis of performance measures for classification tasks', *Information Processing & Management*, 45(4), pp. 427-437. doi: 10.1016/j.ipm.2009.03.002.
23. Song, K., Tan, X., Qin, T., Lu, J. and Liu, T.-Y. (2020) 'MPNet: Masked and permuted pre-training for language understanding', *Advances in Neural Information Processing Systems*, 33, pp. 16857-16867.
24. Wu, R., Wang, H., Chen, H.-T. and Carneiro, G. (2026) 'Deep multimodal learning with missing modality: A survey', *Transactions on Machine Learning Research*.